

NEURAL BELLMAN OPERATORS

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This paper develops Neural Bellman Operators for policy evaluation and feasible improvement in controlled economies. A constructive continuation retains feasible action witnesses and admits policy-identical native and affine-ReLU realizations. Acquired-state feasibility and verified continuation contrasts connect numerical approximation to the cost of the implemented policy. We prove a directed finite-sweep bound that preserves the incumbent at every state and reaches the original Bellman optimum under exact greedy certificates. Signed residual transport makes the comparison computable without a policy-value oracle; explicit work and storage bounds expose its state-cover and horizon dependence. In the original nonlinear investment economy, complete two- and three-date sweeps are evaluated by independent continuous-law cost intervals. A genuinely nonuniform conventional comparator and a controlled approximately-null perturbation catalogue examine representation and verification choices. Own-incumbent gains are identified, while cross-method differences and complete catalogue work are reported without presuming neural dominance. All earlier theory, applications and adverse evidence remain available in the complete development editions.

KEYWORDS: Dynamic programming, Neural approximation, Policy certification, Information acquisition, Computational economics.

1. INTRODUCTION

A continuation value is useful to an economist because it changes a decision. Its prediction error is not the loss of the policy actually implemented. Nor is a smaller upper bound on that loss evidence that one policy has lower cost than another. When observation, action selection and verification are costly, the numerical procedure and the economic decision have to be evaluated together.

We develop Neural Bellman Operators as a policy-evaluation and feasible-improvement method. The economic object remains a controlled economy with policy-specific continuations. The operator constructs each date’s continuation using its own fitted future, returns a feasible policy, and accounts for its loss against the Bellman optimum. A general signed policy sandwich permits arbitrary fitted candidates. A constructive affine-ReLU backend additionally produces the candidate from primitive moduli, without assuming that a nonconvex training algorithm eventually succeeds. Exact compilation retains the original action witness through off-grid evaluation and state-dependent feasibility repair.

The principal question in this revision is how that construction produces a certified economic answer with finite resources. State coverage, action search and integration have different error coefficients. Tying them to one grid size conceals the allocation problem. We derive a separated primitive budget and an optimal continuous allocation for its leading work proxy, including lower resource bounds and dyadic implementation. The empirical comparison then records the complete positive-tolerance frontier of a fixed common resource ladder, not only an isolated favorable target. All failed earlier attempts belong to the work required to return a policy.

The next question is what the returned policy does. We evaluate the actual policies at the declared first crossings, even when two generators cross at different resolutions. Independent interval simulation preserves the original continuous initial and innovation laws and encloses discontinuous actors and robust capacity repair. Conditioning on construction information permits simultaneous cost inference after this selection. The result does not infer a policy ranking from two certificates and does not extrapolate three deterministic timing repetitions into an optimizer-success probability.

Two economic decisions make this distinction operational. First, a controller replacement is worthwhile only if its actual gain exceeds the replacement fee. We identify the directional gain intervals and the full break-even fee account, retaining but no longer relying on a single generous charge. Second, observation precision is chosen on the same finite bit grid on which the implemented policies are evaluated. A simultaneous upper-net-cost selector has a regret bound relative to the best

precision in that catalogue. The bound holds for every nonnegative information price on one coverage event. This is a result about actual net policy cost, different from minimizing a conservative all-state loss account.

These advances do not depend on calling a classical envelope neural. A native min-plus envelope and its exactly equivalent ReLU circuit are one method with two encodings. The original-owner compiler is available to both. We compare that method with uniform multilinear fitted-value iteration (FVI) and two separately recorded conventional refinement rules. The original curvature-bisection block remained uniform; the later graded block was nonuniform. A new residual-driven comparator below records its realized geometry and all pilot work. Each generates its own future labels and receives the same feasible action menu, continuous-law integration and one-sided economic criterion. The evidence must determine the cost of the returned policy as well as certificate sharpness. No uniform neural dominance is built into the design.

1.0.0.1. *Related methods.* Lipschitz extension supplies the cone construction (McShane, 1934), and separable distance transforms supply its linear-work nodal engine (Felzenszwalb and Huttenlocher, 2012). Approximation in min-plus spaces has an established dynamic-programming literature: Lakshminarayanan and Bhatnagar (2014) study min-plus projected Bellman equations and supremum-norm errors. Bellman-inequality methods likewise construct lower value bounds and policies; Wang et al. (2015) use iterated inequalities with linear or semidefinite optimization. Our argument does not claim priority for these principles. Its subject is their connection to an own-future feasible policy, original-owner continuous compilation, acquired-state implementation, complete resource allocation and direct economic comparison of the returned objects. The empirical-Bernstein component uses the bounded-sample inequality of Maurer and Pontil (2009); conditional selection and numerical path enclosures specify the estimand to which it is applied.

The broader NBO program includes controlled diffusions, recursive utility, endogenous preferences, temporal selves and games. Those results and their assumptions are retained in the complete development edition and its full proof supplement, together with the original training, matrix-factor and precision experi-

ments. The present article supplies a self-contained constructive theorem–evidence chain. Its direct-cost statistical theorems concern expected discounted costs, not an undeclared extension to every nonlinear recursive application. This organization preserves the paper’s economic subject while making the current contribution and the actual comparison burden explicit.

The additional improvement result addresses the economically important case where the certified construction returns a more costly policy than a conventional competitor. We preserve that incumbent and permit only actions whose actual reference advantage is certified nonpositive after the numerical and observation allowances. At the final date of the original investment model, exact continuous-law integration makes this gate executable without assuming a successful policy-value fit. Both generators receive the same repair. The resulting experiment isolates removable final-action loss from the policy gap that remains; it does not rename the compiler as a uniquely neural advantage.

The relevant continuation error can be smaller than its global range. We show that a coupled-residual certificate controls the error in feasible action comparisons, and use it in a stronger finite-sweep bound for the same Bellman optimum. A verified nonconstant action-null component need not consume any improvement budget. The original investment dynamics admit an explicit one-unit ReLU example. An exhaustive observation-cell diagnostic holds the returned economic policies fixed while varying this nuisance; it tests certificate robustness, not another expected-cost ranking.

1.0.0.2. From continuation construction to a complete improved policy. The current contribution completes a chain within the same NBO model. The constructive back-end supplies a feasible acquired incumbent; verified signed differences compare that incumbent with implementable proposals; continuous-action covers give a bound against the original feasible infimum; and independent cost intervals evaluate the full policy returned after each pass. Section 7 proves the directed recursion and supplies a streamed, common-innovation realization with explicit resource bounds. Section 14 executes all dates for the full horizon, not merely the last decision. The

earlier final-date and exact-null experiments remain distinct observations rather than being relabeled as full-sweep evidence.

The empirical ordering starts from actual expected policy cost. A complete-catalogue work account includes primitive construction, all executed improvement passes, durable records and the shared cost-inference service. It is an observed cost of returning a policy from a finite catalogue, not an estimate of an unexecuted optimal stopping strategy. Own-incumbent improvement, conventional-method comparison and the robustness of a certificate are reported separately.

1.0.0.3. *Policy iteration, invariance, and state metrics.* Approximate policy iteration already has a substantial error-propagation theory; Munos (2003) studies approximation errors in policy iteration. Conservative policy iteration connects improvement to an incumbent and a performance criterion (Kakade and Langford, 2002), and safe policy iteration develops related monotonic-improvement guarantees (Pirotta et al., 2013). We do not claim the general idea of safe improvement. Our finite-horizon statement requires an acquired-cell feasible action, a verified directed advantage enclosure, and a lower cover of the entire continuous feasible action set. Those conditions identify the error that enters each pass and make the economic cost of verification explicit.

Couplings also occur in bisimulation metrics: Ferns, Panangaden, and Precup (2004) relate state distances to optimal values. Here the transported object is the signed evaluation error of a fixed incumbent, and the pair computation may use any verified coupling with the correct marginals. It is neither a proposed state-aggregation metric nor an assumption that a transport optimum can be obtained without cost. Potential-based reward shaping preserves policies by transforming the reward specification (Ng, Harada, and Russell, 1999); the action-null construction here leaves the economic reward and transition laws unchanged and concerns continuation components with identical feasible-action expectations. Value equivalence instead identifies models through their Bellman updates on specified functions and policies (Grimm, Barreto, Singh, and Silver, 2020). Our contrast equivalence is a statement about continuation errors under the fixed actual kernels. These connections explain the inherited principles and locate the additional contribution

in implementable certification, finite-sweep accuracy and complete policy-cost accounting.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t , the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j < t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v , define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \quad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \quad (1)$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. The direct-cost inference in Section 9 is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It forms targets using its *own* fitted f_{t+1} , returns a neural continuation and a feasible policy, and records an economic error allowance. It does not receive the optimal continuation as a training label. Arbitrary trained networks can be evaluated by the first result below. The subsequent constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. Action-dependent errors and complete-policy loss

Let $H(a)$ be a true current cost and $\hat{H}(a)$ its continuation approximation. If $\hat{a}$ is δ -optimal for $\hat{H}$ on the same feasible action set, then

$$H(\hat{a}) - \inf_a H(a) \leq \delta + \text{osc}_a \{\hat{H}(a) - H(a)\}. \quad (2)$$

Indeed, compare $\hat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \leq u_t, \quad \mathcal{T}_t^\pi f_{t+1} - f_t \leq v_t, \quad \ell_T \leq f_T - g \leq u_T.$$

Then the actual feasible policy satisfies

$$0 \leq J_0^\pi - V_0^* \leq \sum_{t < T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \quad (3)$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t . The terminal width is unchanged. *Q.E.D.*

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \leq K_t^A \|x - z\|_1, \quad L_t = A_t^Q + D_t K_t^A, \quad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t -Lipschitz whenever f_{t+1} is L_{t+1} -Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t , form effective feasible k_t -nets. Evaluate their own-future objectives with certified error at most e_t . Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \leq K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \quad (4)$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \quad (5)$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g .

THEOREM 2—Own-future construction with feasible witnesses: *The functions in (5) have exact affine–ReLU realizations and Lipschitz moduli L_t independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \leq f_t - \mathcal{T}_t f_{t+1} \leq D_t k_t + e_t + 2L_t h_t, \quad (6)$$

$$\mathcal{T}_t^\pi f_{t+1} - f_t \leq e_t + D_t \zeta_t,$$

and hence

$$J_0^\pi - V_0^* \leq \sum_{t < T} B_t(D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T(2e_T + 2L_g h_T). \quad (7)$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t . At a nearest node its cone exceeds the infimum by at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \leq y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \text{ReLU}(z) + \text{ReLU}(-z)$ and $\min(u, v) = u - \text{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. Q.E.D.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are one method with two encodings. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

3.1. Original-owner compilation on a tensor cover

On a tensor cover $\mathcal{X} = \prod_{j=1}^d \{x_{j,0}, \dots, x_{j,n_j}\}$, put $S = \prod_j (n_j + 1)$. For arbitrary labels y_i , including inconsistent inexact labels, compute at each grid node v

$$(z_v, o_v) = \min_{\text{lex}, i} (y_i + L\|v - x_i\|_1, i).$$

The index o_v is the original label owner, not the transformed node v .

THEOREM 3—Continuous original-owner identity: *For a closed tensor cell C with corners $\mathcal{V}(C)$, and $x \in C$,*

$$f(x) = \min_{v \in \mathcal{V}(C)} (z_v + L\|x - v\|_1), \quad (8)$$

$$(f(x), i_*(x)) = \min_{\text{lex}, v \in \mathcal{V}(C)} (y_{o_v} + L\|x - x_{o_v}\|_1, o_v).$$

Coordinatewise forward and backward relaxations compute (z, o) in $O(dS)$ additions and comparisons and $O(S)$ storage. After locating the cell, a query uses $O(d2^d)$ arithmetic operations. Exact rational comparisons preserve all ties, including $L = 0$. Using the original owner’s repaired action preserves the uncompiled policy, its cost and its certificate exactly.

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x ; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$, proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

The nodal algorithm is the classical distance transform (Felzenszwalb and Huttenlocher, 2012); preserving the original action through continuous off-grid queries is the policy obligation. If there are $J_{t,i}$ node actions and M_t integration representatives, write $Q_t = M_t \sum_i J_{t,i}$. Literal dense cone evaluation costs $O(\sum_t dS_{t+1}Q_t)$ in

representation work. Compilation replaces this by $O(\sum_t [dS_{t+1} + Q_t d 2^d])$, plus cell location, primitive evaluation, integration, feasible repair, certificate and output. Rational bit lengths and state boxes crossing many cells incur additional costs. There is no dimension-free complexity assertion.

3.2. Acquired states and conventional actors

A b -bit coordinate sensor reports a dyadic cell and its midpoint $\hat{x}$. On the unit cube, $\|x - \hat{x}\|_1 \leq r(b) = d2^{-b-1}$. For the investment capacity $\kappa(x)$ below, clip the original action to the smallest capacity throughout that cell and round down to an action quantum Δ_t . This is feasible at every state in the cell. Relative to an exact true-state repair, its displacement is at most $2K_t^A r(b) + \Delta_t$. Transporting the witness cone from $\hat{x}$ to x adds at most $2L_t r(b)$, plus any certified selector objective error ν_t . Thus the implemented-policy account adds

$$\sum_{t < T} B_t \{ (2L_t + 2D_t K_t^A) r(b) + D_t \Delta_t + \nu_t \}. \quad (9)$$

No continuity of the discrete owner selector is assumed. At uncertain numerical comparisons the simulator retains all possible actors, rather than choosing a favorable one.

For conventional FVI the continuation is multilinear on its realized tensor grid. Its actual modulus is bounded by the largest coordinate edge slope. A nearest-node actor uses a feasible node minimizer and true-state repair. Let h_t be the largest half-cell ℓ^1 radius, L_t^{gr} the Bellman graph modulus, and e_t the node query allowance. Its optimal residual upper bound is $D_t k_t + e_t + L_t^{\text{gr}} h_t$. Its selected-policy upper bound is $e_t + E_t$, where

$$E_t = \sup_x \{ y_{i(x)} + L_t^{\text{gr}} \|x - x_{i(x)}\|_1 - f_t(x) \}.$$

Partitioning cells by nearest-node boundaries makes the expression separately affine in each coordinate. Its maximum is therefore at a partition vertex; the implementation encloses all such vertices. Both uniform and curvature-refined FVI receive this sharper one-sided account, not an artificially doubled baseline. Their acquired-state allowance is derived separately in (45).

4. COST-DIRECTED FEASIBLE IMPROVEMENT

A policy certificate answers a safety question relative to the optimum. It need not answer the comparison with an installed controller. We now make actual cost reduction a constraint on the improvement step. The continuation representation, including a neural realization, supplies proposals; a centered, policy-specific evaluation determines whether to implement them. The incumbent is always an admissible fallback.

Fix a feasible implemented policy π and bounded functions $h_0, \dots, h_T$ in the common evaluation domain of Section 2. Its observation and rounding rules are part of π . Suppose verified constants satisfy

$$\ell_t \leq \mathcal{T}_t^\pi h_{t+1} - h_t \leq u_t \quad (t < T), \quad \ell_T \leq g - h_T \leq u_T. \quad (10)$$

Define $a_T = \ell_T$, $b_T = u_T$ and

$$a_t = \ell_t + \beta_t a_{t+1}, \quad b_t = u_t + \beta_t b_{t+1}, \quad w_t = b_t - a_t.$$

All inequalities in this section are uniform over the true state domain. Bounds at sampled points alone do not meet this requirement.

PROPOSITION 4—Centered reference advantage: *Under (10), $a_t \leq J_t^\pi - h_t \leq b_t$. For any feasible action v at state x , put*

$$d_t(x, v) = Q_t(h_{t+1}; x, v) - Q_t(h_{t+1}; x, \pi_t(x)).$$

If $\bar{d}_t(x, v)$ is a valid upper enclosure of this difference, then

$$Q_t(J_{t+1}^\pi; x, v) - J_t^\pi(x) \leq \bar{d}_t(x, v) + \beta_t w_{t+1}. \quad (11)$$

The width allowance and the action contrast are invariant to date-specific constant shifts of the fitted continuations.

PROOF: Monotonicity and cash invariance propagate the two error endpoints backward. Applying them once more to each of the two feasible actions bounds each continuation-evaluation error between $\beta_t a_{t+1}$ and $\beta_t b_{t+1}$. Their difference is at

most $\beta_t w_{t+1}$. Date-specific shifts translate both endpoints by the same amount and cancel in the action contrast. *Q.E.D.*

The proposition is deliberately a reference-policy result. Substituting a bound on $h - V^*$ without controlling $h - J^\pi$ changes the statement. Likewise, accepting an action merely because it lowers $Q_t(h_{t+1})$ ignores the reference-evaluation allowance.

THEOREM 5—Incumbent-preserving Bellman improvement: *Suppose a measurable feasible proposal v_t and a nonnegative measurable margin κ_t are given. At a state where a verified enclosure satisfies*

$$\bar{d}_t(x, v_t(x)) + \beta_t w_{t+1} \leq -\kappa_t(x), \quad (12)$$

implement $v_t(x)$; otherwise implement the original $\pi_t(x)$. Call the resulting policy μ . For a cell-based observation rule, require feasibility and (12) for every true state in the reported cell, using the same proposed action on that cell. Then $J_t^\mu(x) \leq J_t^\pi(x)$ at every date and state.

Under conditional expectation, let D_t be the accepted-state set and let the state trajectory follow μ . Then the stronger quantitative assertion is

$$J_0^\mu(x) - J_0^\pi(x) \leq -\mathbb{E}_x^\mu \sum_{t < T} B_t \kappa_t(X_t) \mathbf{1}_{D_t}(X_t). \quad (13)$$

The qualitative assertion holds for the stipulated monotone cash-invariant conditional recursion. The occupancy formula is asserted only for expectation.

PROOF: At accepted states Proposition 4 gives $\mathcal{T}_t^\mu J_{t+1}^\pi \leq J_t^\pi - \kappa_t$. Elsewhere the two actions coincide, so the advantage is exactly zero, not $\beta_t w_{t+1}$. Backward monotone comparison, starting from a common terminal cost, gives the first assertion. Under expectation subtract $J_t^\pi(X_t)$ along the μ trajectory and telescope the conditional expectations. The resulting identity is the expected discounted sum of true reference advantages, each bounded above by the accepted negative margin or by zero. *Q.E.D.*

The improvement principle is related to approximate policy iteration with safeguarded updates, including [Metelli et al. \(2021\)](#). We do not claim that monotone

policy improvement itself is new. Here the relevant connection is between a signed own-policy evaluation account, action-centered numerical errors, original-owner construction, and the actual information-constrained action. The test is available to neural and conventional proposals alike. An exact native encoding of a ReLU continuation does not become a second independent method.

4.1. A fully executed final-date specialization

A useful test of this result leaves every earlier decision fixed and improves only the final action. Set $h_T = g$. The continuation width is then zero, so the gate concerns the actual conditional final cost, without a fitted policy-evaluation error. For the investment economy in Section 11, write

$$y_j = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + \gamma_j a, \quad s_j = (-1)^{j+1}, \quad \sigma_z^2 = \frac{1}{3072},$$

where the cyclic convention and γ_j are as in that section. Put $b = 1/2 - 2\sum_j y_j/d$, $r = |\sum_j s_j|/(16d)$, and

$$\Psi_2(b, r) = \begin{cases} b_+^2, & r = 0, \\ ((b+r)_+^3 - (b-r)_+^3)/(6r), & r > 0. \end{cases} \quad (14)$$

Exact integration under the original continuous innovation law gives

$$\begin{aligned} \mathbb{E}g(y + sZ) &= \frac{4}{d} \sum_j (y_j - 5/8)^2 + \frac{1}{4d} \sum_j (y_j - y_{j+1})^2 + 2\Psi_2(b, r) \\ &\quad + 4\sigma_z^2 + \frac{\sigma_z^2}{4d} \sum_j (s_j - s_{j+1})^2. \end{aligned} \quad (15)$$

Thus no terminal innovation mesh is needed for this improvement step. Its computation is nevertheless charged. The formula includes the odd-dimensional shortage uncertainty; setting it to zero in dimension three would change the model.

COROLLARY 6—Acquired-cell final improvement: *In the declared investment economies, final conditional cost $H(x, a) = c(x, a) + \beta \mathbb{E}g(F(x, a, Z))$ is strictly convex in a , with second derivative at least $2p$. Fix an observation cell C and the implemented*

incumbent action a_C . Choose a lattice minimizer a_C^* at the reported midpoint on the cell-wide feasible set. Accept it only if an outward enclosure of $H(x, a_C^*) - H(x, a_C)$ is nonpositive for all $x \in C$. Keep the incumbent on every other cell and at all earlier dates. The resulting acquired policy weakly improves actual expected discounted cost for every initial state. This guarantee does not compare it to a different full-information incumbent.

The lattice is $\{k/4096 : k \in \mathbb{N}_0\}$, truncated by the minimum capacity on the eight-bit cell. Convex derivative bisection identifies the two neighboring lattice actions; their actual cost contrast selects the midpoint proposal. The gate compares its cost with the incumbent on the entire cell, not just at the midpoint. Exact rational point comparison resolves a numerically ambiguous proposal tie. An uncertain true-state box may intersect several cells; the simulator retains their action hulls rather than selecting a favorable cell.

For each generator, old and repaired policies share their earlier trajectory. Their discounted cost difference therefore equals the conditional final contrast. Centering cancels state-only costs and tightens its interval enclosure. When two generators are compared, their trajectories differ and are simulated separately using common innovations. Both comparisons concern actual policies; neither is obtained by subtracting two policy-loss certificates.

5. FROM SAFE IMPROVEMENT TO FINITE-SWEEP ACCURACY

The preceding improvement result compares an implemented policy with its own incumbent. We now connect that comparison to the original Bellman optimum. This distinction matters economically: a policy can improve strictly while remaining more costly than a conventional alternative. The result below preserves the original objective and gives explicit sufficient conditions for closing its optimality gap; it does not infer those conditions from a small empirical residual.

Consider the finite-horizon expectation model of the paper, with bounded costs, feasible measurable Markov policies, and discount factors $\beta_t \geq 0$, $0 \leq t < T$. Write

$$Q_t(h; x, a) = c_t(x, a) + \beta_t \mathbb{E}[h(X_{t+1}) \mid x, a], \quad (T_t h)(x) = \inf_{a \in A_t(x)} Q_t(h; x, a),$$

and $V_T^* = g$, $V_t^* = T_t V_{t+1}^*$. The benchmark is the original fully observed optimum. An acquired-state policy remains feasible for this benchmark; a restriction to cell-wise actions may, however, impose a positive approximation floor. No vanishing information error is assumed.

Starting from a feasible π^0 , at pass k let h^k have certified signed policy-evaluation bands

$$a_t^k \leq J_t^{\pi^k}(x) - h_t^k(x) \leq b_t^k, \quad w_t^k = b_t^k - a_t^k. \quad (16)$$

These can be obtained from the signed-residual recursion already proved; the approximator need not have a small absolute level error. Let a feasible proposal v_t^k satisfy, uniformly in x ,

$$Q_t(h_{t+1}^k; x, v_t^k(x)) \leq (T_t h_{t+1}^k)(x) + \alpha_t^k. \quad (17)$$

For its centered comparison with the exact incumbent, define $d_t^k(x) = Q_t(h_{t+1}^k; x, v_t^k(x)) - Q_t(h_{t+1}^k; x, \pi_t^k(x))$. A quantitative upper enclosure satisfies

$$d_t^k(x) \leq \bar{d}_t^k(x) \leq d_t^k(x) + \zeta_t^k. \quad (18)$$

Here ζ_t^k includes acquisition-cell variation and integration or arithmetic enclosure width, not just the machine unit roundoff. The new policy uses the proposal exactly when $\bar{d}_t^k(x) + \beta_t w_{t+1}^k \leq 0$ and otherwise uses $\pi_t^k(x)$ unchanged. For an acquired cell, feasibility and this comparison must hold throughout that cell, with a single implementable action and acceptance decision. Bounds (17)–(18) are required on the same cells, including their boundaries.

THEOREM 7—Finite-sweep accuracy with incumbent preservation: *Suppose (16)–(18) hold at each pass, with nonnegative finite α_t^k and ζ_t^k . Put*

$$\epsilon_t^k = \alpha_t^k + \zeta_t^k + 2\beta_t w_{t+1}^k, \quad E_t^k = \|J_t^{\pi^k} - V_t^*\|_\infty.$$

Then $J_t^{\pi^{k+1}} \leq J_t^{\pi^k}$ at every date and state, and

$$E_t^{k+1} \leq \beta_t E_{t+1}^k + \epsilon_t^k, \quad E_T^k = 0. \quad (19)$$

Set $B_{t,0} = 1$ and $B_{t,j} = \prod_{r=t}^{t+j-1} \beta_r$. For $1 \leq m \leq T - t$,

$$E_t^m \leq B_{t,m} E_{t+m}^0 + \sum_{j=0}^{m-1} B_{t,j} \epsilon_{t+j}^{m-1-j}. \quad (20)$$

In particular, if $\epsilon_t^k \leq \bar{\epsilon}_t$ uniformly in k , then after T passes

$$E_t^T \leq \sum_{j=0}^{T-t-1} B_{t,j} \bar{\epsilon}_{t+j}. \quad (21)$$

With exact policy evaluation up to a datewise constant, exact greedy proposals, and exact centered comparisons, the policy is optimal after at most T passes, and no intermediate pass raises its cost.

The hypotheses separate three computational tasks. A trained continuation must be evaluated relative to the policy actually retained; the action search must approximate the original feasible infimum; and the comparison must be valid over the available information set. A guarantee for only one of these tasks does not imply the theorem. The same conclusion applies to native and neural representations when they implement the same objects. In particular, it does not turn their common compilation saving into a neural-specific advantage. Safe policy improvement is an established principle; the result here supplies the signed-width and acquisition-aware error account for the incumbent-preserving NBO construction.

PROPOSITION 8—An implementable cellwise proposal bound: *Let an acquired cell C have representative x_C and radius r_C . Suppose $Q_t(h; \cdot, a)$ is L_t -Lipschitz on the cell for every menu action, and $Q_t(h; x, \cdot)$ is D_t -Lipschitz on feasible actions. Let $M_C \subseteq \bigcap_{x \in C} A_t(x)$ be a nonempty finite menu such that every $a \in A_t(x)$, for every $x \in C$, lies within $\delta_{A,C}$ of some member of M_C . If $v_C \in M_C$ is an η_C -minimizer at x_C , then it satisfies (17) on C with*

$$\alpha_t^k \leq \eta_C + 2L_t r_C + D_t \delta_{A,C}.$$

The directed-cover assumption is substantive when constraints move with the state. It is not implied by a dense grid at the representative alone. A failed cover or

a cell that cannot certify a comparison retains its incumbent; it must not be counted as a successful accuracy refinement.

PROPOSITION 9—Sharpness of the evaluation-width coefficient: *Under the theorem’s assumptions, the coefficient two on $\beta_t w_{t+1}^k$ in (19) cannot be uniformly replaced by a smaller constant, even for a one-period deterministic problem with two actions, an exact greedy proposal, and exact contrast arithmetic.*

The theorem is constructive but conditional on verified error budgets. It does not assert that a continuous neural fit uniformly approximates a discontinuous acquired-policy value, or that refining a fixed acquisition contract eliminates its information floor. The executed investment study below certifies a final-date improvement. It is not an execution of T full improvement passes, nor numerical evidence of the finite-sweep near-optimality conclusion. Exact finite-state regression tests exercise the new identities, rejection cases, and sharpness example separately from the economic benchmark.

6. POLICY-RELEVANT CONTINUATION ERRORS

A continuation error matters for an investment decision through the actions it changes. The signed-width account above removes a datewise level error. It can nevertheless charge for a nonconstant error that has the same conditional expectation under every feasible action. We now obtain a sharper account without changing the Bellman optimum, the acquisition contract, or the incumbent-preserving rule.

Write $P_t^a f(x) = \mathbb{E}[f(X_{t+1}) \mid x, a]$, and at pass k put $e_t^k = J_t^{\pi^k} - h_t^k$. An *action-contrast certificate* is a measurable nonnegative bound $R_t^k(x; a, b)$ satisfying

$$|(P_t^a - P_t^b)e_{t+1}^k(x)| \leq R_t^k(x; a, b) \leq \chi_t^k \quad (a, b \in A_t(x)). \quad (22)$$

Both inequalities are required uniformly over true states and feasible actions. The scalar certificate $R_t^k = w_{t+1}^k$ always remains available. A bound computed for the selected proposal alone is sufficient for safety, but not for the accuracy conclusion, which compares with the entire feasible infimum.

THEOREM 10—Contrast-certified finite-sweep improvement: *In the expectation model of Theorem 7, retain the feasible proposal and comparison-enclosure assumptions (17)–(18). Suppose (22) holds. Accept $v_t^k(x)$ precisely when*

$$\bar{d}_t^k(x) + \beta_t R_t^k(x; v_t^k(x), \pi_t^k(x)) \leq 0; \quad (23)$$

otherwise retain the exact incumbent. On an acquired cell, the proposal, acceptance decision, feasibility, and all enclosures must be valid on the whole cell. Then every pass is nonworsening at every state and date. Moreover,

$$E_t^{k+1} \leq \beta_t E_{t+1}^k + \tilde{\epsilon}_t^k, \quad \tilde{\epsilon}_t^k = \alpha_t^k + \zeta_t^k + 2\beta_t \chi_t^k. \quad (24)$$

The finite-sweep unrolling and T-pass bound of Theorem 7 hold with ϵ replaced by $\tilde{\epsilon}$. In particular, exact greedy search and comparisons with $\chi_t^k = 0$ give an optimal policy after at most T passes, even when the global evaluation widths are positive. The coefficient two remains sharp over this class of certificates.

The theorem concerns expected discounted cost. It does not silently extend the linear action-kernel difference to a nonlinear recursive preference. The earlier monotone-recursion safety results remain in force under their own hypotheses. Nor does a small sampled action contrast establish (22). The next result constructs an upper bound using residuals and couplings with verified marginals; it never calls the unknown policy value.

PROPOSITION 11—Coupled residual certificate: *Fix a policy π and critic h . Let $r_T = g - h_T$ and $r_t = \mathcal{T}_t^\pi h_{t+1} - h_t$, and let w_t be valid signed-residual widths. For every pair of states let $\Lambda_t^\pi(x, y)$ be a measurable coupling of $P_t^{\pi_t(x)}(x, \cdot)$ and $P_t^{\pi_t(y)}(y, \cdot)$. Suppose verified nonnegative functions H_t enclose $|r_t(x) - r_t(y)|$. Define*

$$D_T(x, y) = \min\{w_T, H_T(x, y)\},$$

$$D_t(x, y) = \min\left\{w_t, H_t(x, y) + \beta_t \int D_{t+1}(u, v) d\Lambda_t^\pi(x, y)(u, v)\right\}. \quad (25)$$

Then $|e_t(x) - e_t(y)| \leq D_t(x, y)$. For any measurable coupling $\Lambda_t(x; a, b)$ of the two feasible action kernels,

$$R_t(x; a, b) = \min \left\{ w_{t+1}, \int D_{t+1}(u, v) d\Lambda_t(x; a, b)(u, v) \right\} \quad (26)$$

is a valid action-contrast certificate. Replacing either integral by a verified nonnegative upper enclosure preserves validity. No optimal-transport minimizer or smooth incumbent is required.

The construction includes the work needed to enclose residual differences, transport them, and take the action supremum. It is an alternative certificate, not a free evaluation oracle. In a finite model, a common-uniform coupling gives an explicit terminating implementation. In a continuous model, an available common innovation gives a coupling directly, but integration and cellwise certification still have to be supplied. With coarse enclosures the minimum in (26) returns the original scalar guarantee.

6.1. Action-null components and neural realizations

Define the action-null space at date t by

$$\mathcal{N}_t = \{n : (P_t^a - P_t^b)n(x) = 0 \text{ for every } x \text{ and } a, b \in A_t(x)\}.$$

This is a linear space of bounded measurable functions, not a statement about the accuracy of a training procedure. Constants belong to it; exogenous payoff components can make it larger. Values differing by an element of $\mathcal{N}_t$ have identical action contrasts at date t , although their levels and state dependence need not agree.

COROLLARY 12—Certified removal of action-null error: Suppose $e_{t+1}^k = n_{t+1}^k + u_{t+1}^k$, where membership $n_{t+1}^k \in \mathcal{N}_t$ is established and u_{t+1}^k has a verified range of width v_{t+1}^k . Then $R_t^k \leq v_{t+1}^k$ is available, independently of the amplitude of n_{t+1}^k . A constructive way to obtain this width is to evaluate the corrected critic $h^k + n^k$ by the original signed-residual recursion. The action comparisons using h_{t+1}^k and $h_{t+1}^k + n_{t+1}^k$ coincide.

The qualification “established” is important. Arbitrarily declaring part of a fit to be null invalidates the certificate. If an exact null component is known, symbolic

cancellation before interval arithmetic avoids subtracting two large levels. If it is only approximately null, its verified nonzero action contrast must be added to the allowance.

PROPOSITION 13—A nonconstant ReLU component in the original economy: *In the investment dynamics (48), for each declared dimension $d \geq 2$, every bounded measurable function of $y_1 - 2y_2$ is action-null under the common innovation. In particular,*

$$n_M(y) = M(y_1 - 2y_2 + 1)_+, \quad M \geq 0,$$

is a one-unit ReLU realization with oscillation $2M$ on the state cube. At the final date, the critic $h_T = g - n_M$ has global evaluation width $2M$ but zero action-contrast error. The contrast-aware terminal policy is therefore exactly the unperturbed policy of Corollary 6; its statewise cost guarantee is unchanged for every M .

This result supplies a representation-level robustness property of NBO, not an identified advantage over a native encoding of the same continuation. It also identifies an avoidable source of conservatism in a global critic-error account. It does not remove an economically relevant continuation error, a capacity constraint, or an information floor. Full proofs and an exact finite-state reference implementation are in the supplement.

7. DIRECTED CERTIFICATES AND COMPLETE POLICY IMPROVEMENT

The action-contrast result permits an error component to be large when it is irrelevant to a decision. A second source of conservatism remains: taking an absolute value before comparing two actions discards the direction of a useful continuation difference. We retain that direction throughout the verification calculation. The policy generator, the original economic primitives, and the Bellman optimum are unchanged.

7.1. Whole-cell improvement and accuracy

Fix a complete incumbent π^k and write

$$A_t^k(x, a) = c_t(x, a) + \beta_t P_t^a J_{t+1}^{\pi^k}(x) - J_t^{\pi^k}(x).$$

Thus $A_t^k(x, \pi_t^k(x)) = 0$. Let C be an observation cell. Candidate actions must be implementable and feasible for every true state in C . A finite family $\mathcal{B}_t(C)$ must cover every feasible action at every such state; a covering set may contain additional, infeasible actions if its lower enclosure remains valid on the feasible subset. Suppose verified numbers satisfy

$$\begin{aligned} A_t^k(x, a_i) &\leq u_t^k(C, a_i) & (x \in C), \\ \ell_t^k(C, B) &\leq A_t^k(x, a) & (x \in C, a \in B \cap A_t(x)). \end{aligned}$$

All numerical errors are included in these endpoints. The incumbent is an additional candidate with exact upper endpoint zero. Put

$$U_t^k(C) = \min\{0, \min_i u_t^k(C, a_i)\}, \quad L_t^k(C) = \min\{0, \min_{B \in \mathcal{B}_t(C)} \ell_t^k(C, B)\}. \quad (27)$$

Select an action attaining $U_t^k(C)$, retaining the incumbent on a tie, and implement all selected dates simultaneously. This is a finite computation; it does not presume access to the policy value in the display defining A .

THEOREM 14—Directed finite-sweep certificate: *Consider the bounded, finite-horizon expected-cost model of Theorem 10. Suppose the whole-cell feasibility and enclosure conditions above hold at every pass. Set*

$$\varepsilon_t^k = \sup_C \{U_t^k(C) - L_t^k(C)\}.$$

Then $\varepsilon_t^k \geq 0$, $J_t^{\pi^{k+1}}(x) \leq J_t^{\pi^k}(x)$ at every state and date, and

$$E_t^{k+1} \leq \beta_t E_{t+1}^k + \varepsilon_t^k, \quad E_t^k = \sup_x \{J_t^{\pi^k}(x) - J_t^*(x)\}. \quad (28)$$

Writing $B_{t,0} = 1$ and $B_{t,j} = \prod_{i=0}^{j-1} \beta_{t+i}$, for $K \geq 1$ and $m = \min\{K, T - t\}$,

$$E_t^K \leq \sum_{j=0}^{m-1} B_{t,j} \varepsilon_{t+j}^{K-1-j} + B_{t,m} E_{t+m}^{K-m}.$$

The final term vanishes when $K \geq T - t$. In particular, exact directed greedy certificates yield the original Bellman optimum within T passes. The coefficient one on ε cannot be reduced uniformly.

PROOF: For every $x \in C$, the selected action has advantage at most $U_t^k(C) \leq 0$. Starting from the common terminal value and proceeding backwards, monotonicity of conditional expectation gives $J_t^{\pi^{k+1}} \leq J_t^{\pi^k}$. The covering property also gives

$$A_t^k(x, \pi_t^{k+1}(x)) - \inf_{a \in A_t(x)} A_t^k(x, a) \leq U_t^k(C) - L_t^k(C).$$

The left side is nonnegative, so the right side is nonnegative. Replacing the new continuation by the old one using the already proved safety inequality, and then comparing with an arbitrarily near-optimal Bellman action, yields (28). Taking the limit removes any need for attainment of the continuous-action infimum. Iteration proves the displayed finite-sweep bound. For sharpness, take one decision date, an incumbent of cost $\delta > 0$, and an unproposed feasible action of cost zero. The exact incumbent upper advantage is zero and the exact cover lower advantage is $-\delta$. Retaining the incumbent gives both $E = \delta$ and $\varepsilon = \delta$. *Q.E.D.*

This result does not replace Theorem 10. The earlier theorem is useful when only a symmetric error bound is available. The present certificate is useful when a numerical calculation retains signed endpoints. In either case safety for a proposed action is weaker than accuracy against the full feasible infimum. The action cover in (27) is indispensable for the latter. A small gap on a finite proposal menu alone is not a small Bellman gap.

7.2. Signed residual transport without a policy-value oracle

Fix a critic h and incumbent π . Let $e_t = J_t^\pi - h_t$ and $r_t = \mathcal{T}_t^\pi h_{t+1} - h_t$, with $r_T = g - h_T$. Suppose $\mathcal{H}_t(x, y)$ is an interval enclosing $r_t(x) - r_t(y)$. For a coupling $\Lambda_t^\pi(x, y)$ with the stated policy-kernel marginals, interval addition and integration define

$$\mathcal{D}_T(x, y) = \mathcal{H}_T(x, y), \quad \mathcal{D}_t(x, y) = \mathcal{H}_t(x, y) + \beta_t \int \mathcal{D}_{t+1}(u, v) d\Lambda_t^\pi(x, y)(u, v). \quad (29)$$

Integration means an enclosure of the lower and upper integrals, not midpoint quadrature. Independently valid signed evaluation bands may be intersected with

this interval; an empty intersection signals a failed certificate and must stop acceptance.

PROPOSITION 15—Directed residual enclosure: *The interval $\mathcal{D}_t(x, y)$ encloses $e_t(x) - e_t(y)$. If $\Lambda_t(x; a, b)$ couples the two action kernels and $\mathcal{A}_t^h(x; a, b)$ encloses*

$$c_t(x, a) - c_t(x, b) + \beta_t(P_t^a - P_t^b)h_{t+1}(x),$$

then

$$\mathcal{A}_t^h(x; a, b) + \beta_t \int \mathcal{D}_{t+1}(u, v) d\Lambda_t(x; a, b)(u, v) \quad (30)$$

encloses $A_t^\pi(x, a)$ when $b = \pi_t(x)$. Replacing each integral by an outward enclosure preserves the result. Taking absolute endpoint maxima recovers an admissible symmetric action-contrast certificate.

PROOF: The identity $e_t = r_t + \beta_t P_t^\pi e_{t+1}$ and the marginal conditions imply the corresponding identity for differences at (x, y) . Backward induction gives (29). Substitute $J_{t+1}^\pi = h_{t+1} + e_{t+1}$ into the advantage and use the action-kernel marginal conditions to obtain (30). Endpoint integration is order preserving because coupling weights are nonnegative. Intersection with another valid enclosure preserves containment. *Q.E.D.*

The choice $h = 0$ is a valid verification gauge: $e = J^\pi$, the running residual is the incumbent's stage cost, and the terminal residual is g . It does not change the previously constructed policy or claim that a zero critic is an effective policy generator. Under this gauge, a common-innovation rollout of stage-cost differences computes the signed recursion directly. Its computational cost must therefore be charged as policy evaluation. Neither a neural critic nor a native representation receives that work for free.

7.3. A computable specialization in the original investment economy

The full-sweep implementation uses the same two-dimensional nonlinear investment primitives as the preceding study. For two coupled states x, y , retain a separate enclosure of $\Delta = x - y$ rather than recomputing it by subtracting two large state

boxes. For the cyclic bilinear transition, the deterministic difference in coordinate j is

$$\Delta_j \left(\frac{1}{2} + \frac{1 - x_v}{16} \right) + \Delta_v \left(\frac{1}{8} - \frac{y_j}{16} \right) + b_j(a - b), \quad v \neq j, \quad (b_1, b_2) = \left(\frac{1}{2}, \frac{1}{4} \right).$$

A common innovation cancels in this difference, but is still integrated with its original continuous marginal law. Centered polynomial identities evaluate differences of quadratic and quartic costs. For a shortage term, the intersection

$$u_+ - v_+ \in ([u]_+ - [v]_+) \cap [\min\{0, \underline{\Delta}\}, \max\{0, \bar{\Delta}\}]$$

retains a verified interval for $\Delta = u - v$. In regions where both arguments are positive, their difference is exactly Δ ; where both are negative it is zero. These identities are evaluated outward.

The acquired actor is discontinuous at cell boundaries. Its action range on a rectangle is obtained from exact minima and maxima of integer action indices, including all intersected cells. No Lipschitz assumption is imposed on that actor. Innovation intervals are integrated by enclosures on dyadic bins with their exact probabilities. The final innovation is integrated analytically using the original polynomial and shortage identities. The resulting lower covers include every continuous feasible action, whereas only robustly feasible lattice actions can be selected.

PROPOSITION 16—Work, storage, and certified approximation: *Suppose an acquired actor has n^d cells per date and a range-minimum and range-maximum table over dyadic rectangles. Its preprocessing and stored table size are $O(Tn^d(1 + \log n)^d)$, and a rectangle query uses $O(d + 2^d)$ work. Let N be the number of verified observation cells, A the total number of candidate and lower-cover intervals per cell, and q the number of innovation bins at each unresolved date. If the last innovation is integrated analytically, one full pass visits at most*

$$AN \sum_{r=1}^T \sum_{j=0}^{r-1} q^j \tag{31}$$

pair nodes. Streaming the innovation children requires $O(Tmd)$ working state storage for a batch of m roots, in addition to actor tables and $O(TAN)$ retained endpoint records. Thus a

full pair-state table is unnecessary, but tensor-state and exponential-horizon costs have not disappeared.

More generally, suppose an approximate signed pair calculation has a verified terminal endpoint error η_T and local endpoint error at most η_t at date t , inclusive of residual approximation, compression, integration, and arithmetic. Expanding each computed interval by

$$b_T = \eta_T, \quad b_t = \eta_t + \beta_t b_{t+1}$$

is sufficient to enclose the exact signed recursion. In an action-kernel integral the continuation expansion is multiplied by β_t ; its own integration and arithmetic errors must also be added.

PROOF: There are n^d base locations and at most $(1 + \log_2 n)^d$ side-length combinations. Each rectangular extremum is reconstructed from 2^d overlapping dyadic rectangles. A root with r remaining dates has at most $1 + q + \dots + q^{r-1}$ visited nodes; summing over roots, dates, and action intervals gives (31). Processing one child at a time gives the stated stack bound. Finally, integration against a probability coupling is nonexpansive in a uniform endpoint error. Backward induction then proves the error expansion. Q.E.D.

In the executed case, $d = 2$, $n = 32$, $A = 17$ (nine proposals and eight continuous-action cover intervals), and $q = 16$. Each actor table occupies 149,760 bytes in the recorded integer representation, hence 299,520 bytes for two dates and 449,280 bytes for three. These are table bytes, not process peak memory. Identity pairs can terminate without recursion and reduce actual work below (31). Verified compression could further reduce storage, but its error must enter the displayed recursion; unverified interpolation across policy jumps is inadmissible. The present experiment executes streamed interval recursion, not a low-rank pair approximation.

7.4. Approximately null components and misspecification

An exactly null direction is a special case of a quantitative bound. Let $n(y) = M\sigma(v^\top y + b)$, where σ is one-Lipschitz. For a verified coupling of the actual action

kernels, suppose $s(x; a, b)$ bounds $\mathbb{E}|v^\top(Y^a - Y^b)|$. Then

$$|(P_t^a - P_t^b)n(x)| \leq |M|s(x; a, b). \quad (32)$$

This follows directly from the Lipschitz inequality and integration. For a common-innovation affine exposure $Y^a - Y^b = B(a - b)$, the bound is $|M| |v^\top B(a - b)|$. If $v = v_0 + \delta v$, $B = B_0 + \delta B$, and $v_0^\top B_0 = 0$, the non-null coefficient is

$$\delta v^\top B_0 + v_0^\top \delta B + \delta v^\top \delta B.$$

Any action-dependent innovation discrepancy under the actual coupling contributes its own expected absolute projection. These terms cannot be set to zero because a fitted component resembles an invariant.

The deposited perturbation catalogue varies the ReLU direction, action exposure, and an action-dependent innovation-mean shift. Its positive bound is substituted into the same incumbent gate. It also records the policy that would result from falsely declaring the component null. This is a controlled model-misspecification test with known perturbed primitives, not a learned-null discovery procedure or an inference guarantee for an unknown transition law. The original investment laws and the original cost comparisons remain unchanged.

8. ALLOCATING RESOURCES TO AN ECONOMIC TOLERANCE

A single resolution parameter imposes an economic restriction that the Bellman problem does not impose. State coverage, action search and integration have different error coefficients and different effects on work. Refining all three at the same rate can therefore spend most of the service budget on the wrong source of error. We separate these choices while retaining the continuation, feasible-witness interpretation and all-state comparison developed above.

8.1. A separated primitive budget

Consider the investment family specified in Section 11, with $d \in \{2, 3, 4\}$ state coordinates, discount $\beta = 15/16$, and uniform coordinate grids with N intervals. At a state node use $K + 1$ feasible action fractions. Integrate the common scalar

innovation using M equal-probability bins and their midpoints. The three integers need not agree. Let L_t be the primitive graph modulus of the constructed witness continuation, and let D_t be its action modulus. A downward numerical displacement of the action grid is denoted by δ_a .

The state and action covering radii satisfy $h_t = d/(2N)$ and $k_t \leq 1/(8K) + \delta_a$. The innovation lies in an interval of length $1/16$. Within a bin its expected absolute distance from the midpoint is $1/(64M)$. Since changing the scalar innovation by $z - z'$ changes the next state by at most $d|z - z'|$ in the ℓ^1 norm, a certified numerical query has an error allowance

$$e_t \leq \frac{\beta d L_{t+1}}{64M} + \xi_t. \quad (33)$$

Here ξ_t covers arithmetic, interval evaluation and label rounding. It does not suppress the integration error. The terminal label allowance is e_T .

PROPOSITION 17—Separated-resource policy account: *Under the feasible-witness hypotheses, the returned policy satisfies*

$$0 \leq J^\pi - V^* \leq \frac{a}{N} + \frac{b}{K} + \frac{q}{M} + \eta, \quad (34)$$

where

$$\begin{aligned} a &= d \sum_{t < T} \beta^t L_t + d \beta^T L_g, & b &= \frac{1}{8} \sum_{t < T} \beta^t D_t, \\ q &= \frac{d}{32} \sum_{t < T} \beta^{t+1} L_{t+1}, & \eta &= \sum_{t < T} \beta^t (2\xi_t + D_t \delta_a) + 2\beta^T e_T. \end{aligned} \quad (35)$$

The primitive coefficients are independent of the computed labels. The leading number of innovation evaluations in target formation is

$$T(N+1)^d(K+1)M. \quad (36)$$

The compiled continuation additionally incurs its corner-evaluation and preprocessing costs. Pilot selection, policy extraction, verification, arithmetic bit lengths and output are separate charged components.

PROOF: Insert the state and action radii and (33) into the feasible-witness loss bound. The selected-policy upper account contributes the second copy of e_t , not a second action-search allowance. Collecting the coefficients of N^{-1}, K^{-1}, M^{-1} gives (35). There are $(N + 1)^d$ state nodes, $K + 1$ actions per node and M innovation representatives per action at each nonterminal date. No independence of fitting errors is used. Q.E.D.

The account explains why a common further state rung need not be a diagonal refinement. In the $T = 3, p = 4, d = 2$ cell, the exact primitive coefficients are approximately $a = 51.1569, b = 2.3650$ and $q = 1.0486$. The choice $(N, K, M) = (128, 16, 8)$ has account about $0.6786 + \eta$, below target one when the numerical remainder fits the remaining budget. The diagonal choice $(128, 128, 128)$ is more accurate, but its target-generation innovation count at that same state grid is $129 \cdot 128 / (17 \cdot 8)$ times larger. This is an arithmetic comparison of two declared allocations, not an observation of the unexecuted diagonal service or a runtime ratio. Both witness and conventional FVI receive the separated allocation in the experiment.

8.2. The optimal continuous allocation and its boundary cases

A useful planning problem minimizes the leading product $N^d KM$ subject to (34). This proxy isolates the primitive evaluation term; it is not the entire operation count or the measured wall clock. Write $E = \varepsilon - \eta > 0$ and suppose first that a, b, q are positive and all optimal resources exceed their required lower bounds.

PROPOSITION 18—Error shares and clipped allocation: *The unique interior minimizer of the continuous proxy has*

$$\frac{a}{N} = \frac{d}{d+2}E, \quad \frac{b}{K} = \frac{1}{d+2}E, \quad \frac{q}{M} = \frac{1}{d+2}E. \quad (37)$$

For lower resource bounds $r_i \geq \underline{r}_i > 0$, put $c = (a, b, q)$, $p = (d, 1, 1)$ and $\bar{x}_i = c_i / \underline{r}_i$. If $E < \sum_i \bar{x}_i$, there is a unique positive λ such that

$$x_i = \min\{\bar{x}_i, p_i \lambda\}, \quad \sum_i x_i = E; \quad r_i = c_i / x_i. \quad (38)$$

These resources minimize the constrained continuous proxy. If $E \geq \sum_i \bar{x}_i$, all lower resource bounds suffice. Coordinates with zero error coefficient remain at their resource lower bound. When dyadic lower bounds are used, rounding each optimal resource upward to a power of two preserves the error guarantee and multiplies this proxy by less than 2^{d+2} .

PROOF: Set $x_i = c_i/r_i$. Minimizing the resource product is equivalent to maximizing $\sum_i p_i \log x_i$ over $0 < x_i \leq \bar{x}_i$ and $\sum_i x_i \leq E$. Strict concavity and the multiplier conditions give p_i/x_i equal on unconstrained coordinates and larger on capped coordinates. This is precisely (38). The sum of its right sides is strictly increasing until all caps bind. The unconstrained case yields (37). Upward dyadic rounding never enlarges an error term and multiplies $N^d KM$ by less than the stated factor. Neither argument bounds rational-arithmetic bit cost or computer time. *Q.E.D.*

The optimization is a standard separable resource-allocation calculation. Its role here is to expose the error coefficients of the actual own-future policy construction, including integration and implementation, rather than to claim novelty for the multiplier argument. A finite experimental menu need not contain the continuous optimum. The prototype uses a common, frozen dyadic menu for all generators and records all of it. It does not describe this menu as a globally optimal tuning policy for either FVI or the witness method.

8.3. A complete tolerance frontier

For method m , let a prospectively fixed ladder return certificates $g_{m,r}$ and cumulative costs $w_{m,r}$, including failed earlier rungs. Define

$$r_m(\varepsilon) = \min\{r : g_{m,r} \leq \varepsilon\}, \quad W_m(\varepsilon) = w_{m,r_m(\varepsilon)}. \quad (39)$$

An empty set is a capped failure, not an interpolated success. Sort the finite union of all certificate values and zero. On each left-closed, right-open interval between successive values, every method's first crossing and complete work are constant. Evaluating the first-crossing rule at the left endpoint therefore reconstructs the entire positive-tolerance frontier exactly, even when successive certificates are not

monotone. An unbounded final interval is retained. Repetitions have separate costs, while their mathematical checkpoints must be checked for identity.

This finite partition supplies the appropriate interpretation of a threshold advantage. At a fixed resolution the witness method may cost more. At a tolerance where it crosses one rung earlier, the smaller certificate may more than offset that cost. Whether this happens, and over which tolerance interval, is determined by (39); it is not determined by a favorable single threshold. Under an additional homogeneous model $g_m(n) = A_m/n$ and $w_m(n) = C_m n^\kappa$, the continuous frontier ratio is $(C_W/C_F)(A_W/A_F)^\kappa$. This last expression is a conditional explanation, not a fitted scaling law for the experiments. The release supplies the exact finite partition, sensitivity around one, complete prefixes and separate fixed-resolution observations.

9. FROM RETURNED POLICIES TO ECONOMIC RESOURCE DECISIONS

An all-state upper bound and a comparison of actual policy costs answer different economic questions. We now connect the policies returned by a construction frontier to direct cost inference and then to a choice of implementation resources. The construction data select the policies; independently generated evaluation paths assess them. This separation permits a valid comparison of first-crossing policies with different resolutions.

9.1. Simultaneous inference after construction

Let $\mathcal{H}$ denote all construction information: primitive inputs, numerical labels, fitted continuations, returned policies and complete certificate ladders. Conditional on $\mathcal{H}$, fix a finite catalogue of implemented-policy contrasts. Each implementation includes its observation rule, original witness or nearest-node selector, true-state feasible repair and action quantization. For a contrast i , let D_i be the difference in expected discounted costs under a specified initial law. A common-innovation path score evaluates the two policies on their own endogenous trajectories. It is not the difference between two regret bounds.

Suppose the initial and innovation uniforms are partitioned into equal bins. Independent uniform bin indices are drawn, and interval simulation encloses the score for every continuous realization inside the drawn bins. All unresolved state,

selector and repair branches remain in the enclosure. If $[A_i, B_i]$ is a deterministic support interval, clip the path endpoints to that support. Write $\underline{m}_i, \bar{m}_i$ for outward sample-mean bounds on the lower and upper endpoint samples and $v_{i,L}, v_{i,U}$ for upper bounds on their unbiased sample variances. For $n \geq 2$ define

$$R_i(v) = \sqrt{2v\ell/n} + \frac{7(B_i - A_i)\ell}{3(n-1)}, \quad \ell \geq \log(4K/\alpha), \quad (40)$$

where K bounds the number of sampled contrasts.

THEOREM 19—Construction-conditional economic intervals: *Under the stated independent-bin model, conditional on any valid construction history $\mathcal{H}$, with probability at least $1 - \alpha$ all catalogue contrasts satisfy*

$$D_i \in [l_i, u_i] := [\max\{A_i, \underline{m}_i - R_i(v_{i,L})\}, \min\{B_i, \bar{m}_i + R_i(v_{i,U})\}]. \quad (41)$$

The same coverage holds unconditionally. Subsequent selection among these covered contrasts, or evaluation at any deterministic implementation-price vector, needs no additional union bound. This conclusion applies only to policies and initial laws included in the catalogue; it does not create direct-cost observations for every unstudied tolerance or initial state.

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i , and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of [Maurer and Pontil \(2009\)](#) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every $\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

In particular, selecting the first policy to meet a specified certificate is permitted when that selection uses construction records rather than the subsequent evaluation paths. Repeated construction clocks are not independent draws from a neural-optimizer population. The statistical randomness here belongs to the evaluation

design. A fixed pseudorandom stream implements a reproducible experiment; it is not itself a proof of the independent-bin assumption.

9.2. Replacement-fee identification

Write $D = J^W - J^F$. If witness is installed, replacing it by FVI changes total cost by $k - D$, where $k \geq 0$ is the replacement fee. If FVI is installed, the change is $k + D$. Thus the two directional gains before fees are D and $-D$.

COROLLARY 20—The break-even fee frontier: *On an event where $D \in [l, u]$, the two directional gains belong to $[l, u]$ and $[-u, -l]$. The unknown absolute cost difference satisfies*

$$|D| \in [\text{dist}(0, [l, u]), \max\{|l|, |u|\}]. \quad (42)$$

For every $k \geq \max\{0, u, -l\}$ neither replacement strictly lowers expected cost. By contrast, $k \geq \max\{0, l, -u\}$ only removes a certified strictly profitable replacement. The two thresholds generally differ. Strict inequalities in the first criterion certify that both replacements strictly increase cost.

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (42). The assertions hold simultaneously for every nonnegative fee on the original coverage event. *Q.E.D.*

The fee frontier is more informative than a conclusion at one generous charge. The earlier 1/64 replacement decision is retained with its original interpretation and confidence account. The new analysis reports the two directional gain intervals and both thresholds instead of selecting a favorable fee after seeing the data. There is no welfare calibration hidden in the units: a decision maker must supply the economically relevant charge, or use the full frontier to see what remains unidentified.

9.3. Observation costs on the same bit grid

Let $\mathcal{B} = \{4, \dots, 16\}$ be the available bit precisions. For one fixed returned controller, write J_b for the expected cost of its actual b -bit, robustly repaired and quantized implementation. Use $b_0 = 16$ as a common reference and obtain simultaneous intervals for $D_b = J_b - J_{b_0}$. The identity interval at $b = b_0$ is exactly $[0, 0]$. Let $s_b = db \sum_{t < T} B_t$ be the discounted number of coordinate-bits purchased; the net cost is $J_b + \lambda s_b$.

THEOREM 21—Certified choice of actual net implementation cost: *Suppose all $D_b \in [l_b, u_b]$ simultaneously. For any $\lambda \geq 0$ define*

$$L(\lambda) = \min_b \{l_b + \lambda s_b\}, \quad U(\lambda) = \min_b \{u_b + \lambda s_b\}, \quad (43)$$

$$\hat{b}(\lambda) = \min \arg \min_b \{u_b + \lambda s_b\}.$$

Then, simultaneously for all $\lambda \geq 0$,

$$0 \leq J_{\hat{b}} + \lambda s_{\hat{b}} - \min_{b \in \mathcal{B}} (J_b + \lambda s_b) \leq U(\lambda) - L(\lambda). \quad (44)$$

The lower and upper functions are finite piecewise-affine envelopes. All changes in their active lines occur at nonnegative intersections of catalogue lines, with exact comparisons deciding ties. The smallest selected precision $\hat{b}(\lambda)$ is weakly decreasing in λ .

PROOF: Subtract the common J_{b_0} . The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ . A minimum of finitely many affine functions changes its active affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\hat{b}(\lambda_2)} - s_{\hat{b}(\lambda_1)}) \leq 0$. Since s_b increases strictly with b , the precision comparative static follows. *Q.E.D.*

This result makes a statement about *actual* net cost within the declared bit catalogue. It differs from minimizing a uniform all-state loss bound. Both criteria are useful: the latter gives distribution-free protection relative to the full-information

optimum, while (44) uses a specified initial law to compare feasible implementations of one returned policy. Neither criterion gives an optimum over arbitrary sensor technologies or all policies.

For a fair comparison the conventional actor receives its own acquisition allowance. Suppose its nearest-node policy has a certified selected-policy account at a reported state, obtained from the graph modulus L_t^{gr} , and its actual interpolated critic has modulus L_t^f . Transporting the same selected node action to the true state, then using cell-wide capacity repair, adds at most

$$(L_t^{\text{gr}} + L_t^f + 2D_t K_t^A) r(b) + D_t \Delta_t. \quad (45)$$

Indeed, transport the feasible node pair directly to the true state, bound its node distance by the reported-state distance plus $r(b)$, and bound $f_t(\hat{x}) - f_t(x)$ by $L_t^f r(b)$. Robust repair adds $D_t(2K_t^A r(b) + \Delta_t)$. This proves (45) without assuming actor continuity. For a witness critic $L_t^f = L_t^{\text{gr}}$, it reduces to the preceding witness allowance. The empirical comparison therefore evaluates the all-state-account minimizer, the interval upper-net-cost selector, and a descriptive midpoint selector on the *same* integer bit and price grid. The midpoint selector is not called the true optimum.

10. JOINT POLICY-COST AND RESOURCE DECISIONS

Let a finite catalogue $\mathcal{P}$ contain implemented policies, including incumbents, repaired policies and conventional alternatives. Construction, verification and direct evaluation are distinct charged activities. Denote their measured deployment-relevant resource vector by r_j ; the published timing convention must specify whether an evaluation shared by several candidates is a common fixed charge or an attributable marginal charge. In particular, a joint paired evaluation cannot be counted as zero or silently charged only to the preferred generator.

Conditional on construction information, suppose a simultaneous event satisfies $J_j \in [L_j, U_j]$ for every $j \in \mathcal{P}$. For nonnegative prices θ , let $k_j(\theta)$ be a known implementation charge, for example $\theta \cdot r_j$ plus an information fee. Define

$$\hat{j}(\theta) \in \arg \min_{j \in \mathcal{P}} \{U_j + k_j(\theta)\}, \quad R(\theta) = U_{\hat{j}} + k_{\hat{j}}(\theta) - \min_{j \in \mathcal{P}} \{L_j + k_j(\theta)\}. \quad (46)$$

A fixed lexicographic tie rule makes the selection measurable.

PROPOSITION 22—Actual net-cost frontier: *On the simultaneous event, for every nonnegative price vector, including one selected after observing the intervals,*

$$0 \leq J_{\hat{j}} + k_{\hat{j}}(\theta) - \min_j \{J_j + k_j(\theta)\} \leq R(\theta).$$

A proposed replacement of incumbent i by j is certified nonworsening in net cost whenever a simultaneous upper bound on $J_j - J_i$, plus $k_j(\theta) - k_i(\theta)$ and the replacement fee, is nonpositive. Otherwise a rule retaining i remains safe. Policy-specific certificate widths can be displayed alongside this frontier, but do not determine its ordinate.

PROOF: Bound the selected true cost above by its upper line and the best catalogue cost below by the minimum lower line. All inequalities hold on the same finite event for every price, so a new price query does not consume another probability allocation. The replacement assertion applies the paired upper bound directly. *Q.E.D.*

Absolute intervals can be considerably wider than centered paired differences. For a decision between an incumbent and its final repair, the centered interval is therefore the appropriate replacement input. A statistically unresolved interval is not evidence of equivalence. In contrast, Corollary 6 supplies a deterministic nonincrease, although it does not necessarily certify a strictly positive improvement large enough to cover an implementation fee.

10.1. Information technology and the price of conservatism

An explicit theoretical observation technology gives the bit price units. Suppose a device reports a dyadic cell in each coordinate at date t . With b bits per coordinate it transmits db bits and requires τdb resource units, where τ is a technological conversion factor. At unit resource price w , and with a setup cost K_b , its present cost is

$$C_b(w, \tau) = K_b + w\tau db \sum_{t < T} B_t. \quad (47)$$

Each reported cell entails its own feasible action contract. In the retained sensor experiment, $K_b = 0$ and the normalized price $\lambda = w\tau$ takes the declared values. This supplies a structural interpretation and a sensitivity variable, not an estimate of

actual device costs. The dataset contains no empirical measurements from which w or τ could be calibrated.

Theorem 21 and Proposition 22 apply to the finite bit catalogue with charges (47). Their upper-minus-lower envelope is a valid bound on the cost of conservative choice relative to the unknown true catalogue minimizer. Replacing that minimizer by the lowest observed sample mean and calling it the ex post true optimum is not justified. The retained exact price envelopes and their regret bounds answer the identified decision question without making this substitution. A richer technology with fixed charges, nonlinear bandwidth costs or precision-dependent delays can be assessed by changing the known charge functions, but new unobserved policies require new cost bounds.

A similarly explicit interpretation applies to terminal repair. A controller that incurs cost K to install the additional gate, and q per deployed final action, is worth adopting only when its expected discounted policy gain covers those charges. The published centered gain interval defines the entire break-even range. Its mathematical safety does not by itself imply economic adoption at positive computation prices.

11. CERTIFIED COMPUTATION AND ECONOMIC DECISIONS

11.1. *The controlled investment family*

There are $d \in \{2, 3, 4\}$ capital coordinates on $[0, 1]^d$, one investment action, and a common continuous shock. The feasible set is

$$0 \leq a \leq \kappa(x) = \frac{1}{8} + \frac{1}{8d} \sum_{j=1}^d x_j.$$

With cyclic coordinate indices, $\gamma_j = 1/2$ for odd j and $1/4$ for even j , and $z \sim \text{Unif}[-1/32, 1/32]$, the dynamics are

$$F_j(x, a, z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + \gamma_j a + (-1)^{j-1} z. \quad (48)$$

Current costs are

$$c(x, a) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2 \left(\frac{1}{2} - \frac{2}{d} \sum_j x_j \right)_+^2 + pa^2 + 4a^4. \quad (49)$$

Terminal cost doubles the first coefficient and retains the imbalance and shortage terms, with no action cost. Capital enters both adjustment incentives and feasible capacity. The nonlinear transition and shortage penalty couple the coordinates; the model is not separable linear-quadratic control. Prices $p \in \{1, 4\}$ and horizons $T \in \{2, 3\}$ are theoretical primitives, not calibrated estimates.

For the declared dimensions, valid primitive moduli in ℓ^1 are

$$C_x = \frac{13}{2d}, \quad L_g = \frac{9}{d}, \quad F_x = \frac{11}{16}, \quad F_a = \sum_j \gamma_j, \quad C_a = \frac{p}{2} + \frac{1}{4}, \quad K^A = \frac{1}{8d}.$$

The transition is invariant by direct interval bounds. The state column sums of its Jacobian are at most $11/16$. The costs have piecewise-affine gradients; the cube vertices and the shortage-boundary vertices give the stated bounds. The supplemental proof restricts this explicit formula to the declared dimensions rather than presuming it for an arbitrary number of coordinates. The general constructive theorem uses whichever valid primitive moduli the economy supplies.

11.2. Matched services and realized adaptation

The execution-before protocol fixes three generators: compiled witness, uniform multilinear FVI, and curvature-refined coordinate FVI. Every generator forms its own future labels and receives the same state-node count, feasible action fractions and integration bins. For each two-state price-horizon cell the common resource ladder is

$$(N, K, M) = (4, 2, 1), (8, 2, 1), (16, 4, 2), (32, 4, 2), (64, 8, 4), (128, 16, 8).$$

The entire ladder is run in three isolated processes per cell and method, rotating method order. This yields 36 two-state services and 216 rungs. State dimension three uses four declared rungs through $(32, 8, 4)$; dimension four uses three through

(16, 8, 4), with the witness and uniform generators, both horizons, price one and three repetitions. Those additional 24 services contain 84 rungs. The comparison across dimensions uses the same target five, not a dimension-specific certificate threshold. This is a finite common-accuracy experiment at a coarse tolerance, not evidence of an asymptotic or dimension-free rate.

The curvature comparator obtains an own-future pilot on a nine-node coordinate grid. For each axis it takes maxima of adjacent exact second differences across the other coordinates. It then bisects the interval maximizing squared length times the largest pilot curvature over that interval, with deterministic leftmost ties. A negligible positive floor permits a defined rule when the pilot is affine; the dominant unit baseline of the earlier heuristic is absent. Pilots, allocation and the realized axes are stored and charged. Realized nonuniformity, rather than the method's name, determines whether the experiment supplies an adaptive comparison. This remains a coordinate tensor algorithm; it is not presented as a sparse-grid or general adaptive-cell method.

Each rung starts from primitives. The internal complete-prefix clock includes own-future integration, all earlier failed rungs, pilots, exact compilation, actor construction, all-state certification, deployment checks and durable checkpoints. Separate process clocks include startup and the identical non-economic warm-up. A single CPU is selected, numerical-library threads are fixed to one, and CPU frequency is not controlled. Peak resident memory includes the process. Primitive query counts are component counts, not complete FLOPs or bit-operation counts. Independent policy evaluation has a separate joint service and is never added to an unrelated historical clock.

TABLE I: Block A separated-resource attainment

Method	4	2	1	1/2
W	12/12	12/12	12/12	6/12
F	12/12	12/12	12/12	3/12
A	12/12	12/12	12/12	3/12

W: compiled witness; F: uniform FVI; A: pilot-curvature bisection FVI. Twelve services per method are four economies, each repeated three times. They are not twelve independent economic tasks or optimizer draws.

TABLE II: Complete positive-tolerance frontiers in two states

(T, p)	Tolerance interval	W: N ; sec.	F: N ; sec.	A: N ; sec.
(2,1)	[0,0.423176)	–	–	–
(2,1)	[0.423176,0.483393)	128; 10.658	–	–
(2,1)	[0.483393,0.846352)	128; 10.658	128; 8.458	128; 9.817
(2,1)	[0.846352,0.962257)	64; 1.458	128; 8.458	128; 9.817
(2,1)	[0.962257,1.6927)	64; 1.458	64; 1.134	64; 1.514
(2,1)	[1.6927,1.90651)	32; 0.290	64; 1.134	64; 1.514
(2,1)	[1.90651,2.79584)	32; 0.290	32; 0.269	32; 0.399
(2,1)	[2.79584,3.21615)	16; 0.092	32; 0.269	32; 0.399
(2,1)	[3.21615,5.59167)	16; 0.092	16; 0.119	16; 0.176
(2,1)	[5.59167,6.13235)	8; 0.034	16; 0.119	16; 0.176
(2,1)	[6.13235,10.0042)	8; 0.034	8; 0.061	8; 0.090
(2,1)	[10.0042,10.6876)	4; 0.014	8; 0.061	8; 0.090
(2,1)	[10.6876, ∞)	4; 0.014	4; 0.029	4; 0.042
(2,4)	[0,0.45093)	–	–	–
(2,4)	[0.45093,0.506837)	128; 10.689	–	–
(2,4)	[0.506837,0.901859)	128; 10.689	128; 8.453	128; 9.797
(2,4)	[0.901859,1.00914)	64; 1.452	128; 8.453	128; 9.797
(2,4)	[1.00914,1.80372)	64; 1.452	64; 1.134	64; 1.509
(2,4)	[1.80372,2.00023)	32; 0.290	64; 1.134	64; 1.509
(2,4)	[2.00023,2.92224)	32; 0.290	32; 0.268	32; 0.397
(2,4)	[2.92224,3.31506)	16; 0.092	32; 0.268	32; 0.397
(2,4)	[3.31506,5.84447)	16; 0.092	16; 0.118	16; 0.177
(2,4)	[5.84447,6.33036)	8; 0.034	16; 0.118	16; 0.177
(2,4)	[6.33036,10.3185)	8; 0.034	8; 0.061	8; 0.091
(2,4)	[10.3185,10.9232)	4; 0.014	8; 0.061	8; 0.091
(2,4)	[10.9232, ∞)	4; 0.014	4; 0.029	4; 0.043
(3,1)	[0,0.63562)	–	–	–

	(T, p)	Tolerance interval	W: N ; sec.	F: N ; sec.	A: N ; sec.	
1						1
2						2
3	(3,1)	[0.63562, 0.690808)	128; 15.635	–	–	3
4	(3,1)	[0.690808, 1.27124)	128; 15.635	128; 12.420	128; 14.176	4
5	(3,1)	[1.27124, 1.37488)	64; 2.070	128; 12.420	128; 14.176	5
6	(3,1)	[1.37488, 2.54248)	64; 2.070	64; 1.651	64; 2.148	6
7	(3,1)	[2.54248, 2.72357)	32; 0.401	64; 1.651	64; 2.148	7
8	(3,1)	[2.72357, 4.11404)	32; 0.401	32; 0.388	32; 0.568	8
9	(3,1)	[4.11404, 4.53322)	16; 0.126	32; 0.388	32; 0.568	9
10	(3,1)	[4.53322, 8.22807)	16; 0.126	16; 0.171	16; 0.252	10
11	(3,1)	[8.22807, 8.6154)	8; 0.046	16; 0.171	16; 0.252	11
12	(3,1)	[8.6154, 14.5143)	8; 0.046	8; 0.088	8; 0.130	12
13	(3,1)	[14.5143, 14.9762)	4; 0.019	8; 0.088	8; 0.130	13
14	(3,1)	[14.9762, ∞)	4; 0.019	4; 0.041	4; 0.061	14
15	(3,4)	[0, 0.678554)	–	–	–	15
16	(3,4)	[0.678554, 0.723386)	128; 15.815	–	–	16
17	(3,4)	[0.723386, 1.35711)	128; 15.815	128; 12.435	128; 14.257	17
18	(3,4)	[1.35711, 1.44001)	64; 2.085	128; 12.435	128; 14.257	18
19	(3,4)	[1.44001, 2.71422)	64; 2.085	64; 1.643	64; 2.169	19
20	(3,4)	[2.71422, 2.85349)	32; 0.402	64; 1.643	64; 2.169	20
21	(3,4)	[2.85349, 4.31287)	32; 0.402	32; 0.390	32; 0.569	21
22	(3,4)	[4.31287, 4.66727)	16; 0.127	32; 0.390	32; 0.569	22
23	(3,4)	[4.66727, 8.62574)	16; 0.127	16; 0.170	16; 0.253	23
24	(3,4)	[8.62574, 8.88407)	8; 0.046	16; 0.170	16; 0.253	24
25	(3,4)	[8.88407, 15.0203)	8; 0.046	8; 0.088	8; 0.131	25
26	(3,4)	[15.0203, 15.306)	4; 0.019	8; 0.088	8; 0.131	26
27	(3,4)	[15.306, ∞)	4; 0.019	4; 0.042	4; 0.061	27

Each time is the median complete prefix over three isolated repetitions. A dash is a capped failure. Adjacent bands with identical returned rungs are merged. Printed cutoffs are rounded; exact rational endpoints in the deposited frontier determine membership, including equality. The action and integration budgets are uniquely fixed by N in the common ladder. Minima, maxima and counts are deposited, not replaced by these medians.

TABLE III: Repeated common-accuracy comparison: target five, price one

d	T	W pass	N	sec.	F pass	N	sec.
2	2	3/3	16	0.092	3/3	16	0.119
2	3	3/3	16	0.126	3/3	16	0.171
3	2	3/3	16	2.424	3/3	16	2.074
3	3	3/3	16	3.508	3/3	16	3.032
4	2	3/3	16	58.569	3/3	16	60.809
4	3	3/3	16	82.156	3/3	16	90.099

All dimensions face the same coarse target five; caps and allocations remain explicit. Two-state observations are reused from the main catalogue and not counted twice. This finite comparison does not identify an asymptotic scaling law.

The original block A contains 60 services and 300 rungs. Repeated mathematical checkpoints are byte-identical. The pilot-curvature bisection generator realizes nonuniform axes in 0 of 252 date models; its name alone does not establish effective adaptation. At common successful declared target-by-repetition comparisons, witness is faster than uniform FVI in 13 of 39 and slower in 26. Against pilot-curvature bisection FVI, the corresponding counts are 30 faster and 9 slower out of 39. These counts summarize selected targets; Table II and the exact deposited partition, rather than a binary attainment headline, are the primary work comparison.

The new first-crossing direct-cost intervals identify lower witness cost in 0 cases, higher cost in 93, and leave 3 unresolved. For the separately evaluated favorable R48 work pair, the corresponding counts are 0, 0 and 8. A cheaper first crossing and a sharper bound therefore receive an actual economic comparison rather than an inferred ranking. The detailed intervals and initial actions are retained, including every adverse outcome.

Across the 24 declared controller-price decisions, the upper-net-cost selector is certified strictly cheaper than the all-state-bound minimizer in 24. Its cost-regret upper bounds relative to the best evaluated bit range from 0.00238686 to 0.00332242 in the normalized economic units. Where the comparison does not identify a strict

improvement it remains unresolved; midpoint choices are descriptive, not true optima. This is a finite expected-net-cost conclusion under the uniform initial law, not a calibrated welfare result or a universal preference for coarse observation.

The direct and sensor study contains 200 sampled contrasts, totaling 52,428,800 interval paths, with a separately recorded joint evaluation time of 923.64 seconds. Identity rows use no simulated paths. This evaluation work is not allocated to one favored generator or hidden inside another method’s construction clock.

11.3. *Actual first-crossing policies and replacement fees*

For the four two-state economic cells and each target 4,2,1, the protocol selects repetition zero’s first-crossing witness and uniform FVI policies when both attain. It also selects the particular favorable R48 target-two pair: $T = 3, p = 4$, witness at $N = 32$ and FVI at $N = 64$. Those old frozen policies are evaluated as an additional pair; their historical clocks are not reconstructed as part of the new service. Each selected pair is compared under the uniform initial law and point laws at $(1/8, 1/8)$, $(1/2, 1/2)$ and $(7/8, 7/8)$, with full information or eight-bit observations. Finite observations use robust cell-wide capacity repair and downward action quantum $1/4096$.

Every sampled contrast uses 262,144 common interval paths, 40-bit equal initial and innovation bins, and outward mean and variance bounds. All new policy-pair and sensor contrasts share one family of at most 300 with error $1/100$ and log upper bound 13. The independent-bin model supplies the statistical assertion; the fixed PCG64 streams supply reproducibility. Old R48 intervals retain their separate original confidence event. Simultaneous interpretation of both old and new families is at least 98 percent by a union bound, not automatically 99 percent.

TABLE IV: Actual costs of the returned policies and fee sensitivity

Pair	ε	Bits	n	W lower	W higher	Unres.	Sufficient fee range
R49	4	E	16	0	14	2	[0.00156, 0.1042]

Pair	ϵ	Bits	n	W lower	W higher	Unres.	Sufficient fee range
R49	4	8	16	0	15	1	[0.001565,0.1047]
R49	2	E	16	0	16	0	[0.007903,0.05741]
R49	2	8	16	0	16	0	[0.007853,0.05892]
R49	1	E	16	0	16	0	[0.003374,0.02665]
R49	1	8	16	0	16	0	[0.003379,0.02688]
R48	2	E	4	0	0	4	[0.001746,0.001875]
R48	2	8	4	0	0	4	[0.001746,0.001886]

E denotes full information. Counts cover every declared initial law and economic cell in the group; they do not rank all initial states. The fee is sufficient to certify that neither directional switch pays. It is not the smaller threshold that merely removes identified profitability. All new intervals share one 99 percent family conditional on the independent-bin model.

The complete numerical supplement reports the directional replacement gains for every pair and the fee sufficient to certify that neither switch pays. It also reports the lower threshold that merely removes an identified profitable switch. The old $1/64$ decision is preserved, but no conclusion depends on treating that fee as calibrated. The deposited exact fee frontiers permit any nonnegative fee to be assessed on the same confidence event. Lower regret certificates are never described as observed policy-cost improvements.

11.4. *Purchasing information on the evaluated grid*

For each first-target-one controller, the protocol evaluates every integer precision $b = 4, \dots, 16$ under the uniform initial law. Each policy is compared with its own 16-bit implementation; the reference self-comparison is exactly zero. Prices per coordinate-bit are $1/16384, 1/4096, 1/1024$. The deterministic all-state-account minimizer and the upper-net-cost selector of Theorem 21 use precisely that same grid. Neither is compared with an unobserved continuous optimum.

TABLE V: Information choice on the same evaluated bit grid

$T;p$	Method	Bound-minimizing bits	Net-selected bits	Net-regret bound range
2;1	W	16,14,12	6,6,4	[0.002563,0.002596]
2;1	F	16,14,12	7,4,4	[0.002387,0.002426]
2;4	W	16,14,12	6,6,4	[0.002691,0.00272]
2;4	F	16,14,12	4,4,4	[0.002564,0.002564]
3;1	W	16,14,12	7,4,4	[0.003092,0.003107]
3;1	F	16,14,12	6,4,4	[0.002992,0.003036]
3;4	W	16,15,13	4,4,4	[0.003322,0.003322]
3;4	F	16,14,12	4,4,4	[0.003237,0.003237]

Bit triples correspond, in order, to prices $1/16384, 1/4096, 1/1024$ per coordinate-bit. The net selector minimizes the upper actual-net-cost interval. Its regret bound is against the best bit in the finite catalogue, not an estimated unconstrained optimum. Every price-specific value and all nonnegative-price envelopes are deposited.

For each economic cell, the record additionally gives all nonnegative information prices by exact lower and upper affine envelopes. A retrospective price query within that family is allowed by simultaneous coverage and is not a new experiment. Initial-law dependence remains substantive: a uniform-state expected-cost decision need not be optimal at a particular initial state, and a conservatively optimal all-state bound may rationally purchase more information.

11.5. *The retained evidence and its interpretation*

The new service does not rewrite the original nonlinear training studies. The R42 tighter-target neural attainments, six of twelve scalar and five of six coupled services, remain capped-service outcomes; the R44 residual refinements of seven frozen failures remain diagnostics, not retraining successes. The 24 direct neural-minus-ridge intervals remain unresolved. The R45 constructive comparison still has nine of twelve neural and twelve of twelve spline attainments at its tightest target.

Fixed64 and adaptive precision still certify all 36 historical precision services, with adaptive slower in 26 and about 0.98 percent slower in aggregate.

The R48 target-one count, twelve of twelve for witness versus six of twelve for each FVI method, also remains unchanged. At a relaxed tolerance 1.014 the old FVI near misses all attain; the release supplies a complete sensitivity table rather than reclassifying those original failures. The new common $N = 128$ state rung uses separately declared action and integration budgets. It is not the unexecuted diagonal $(128, 128, 128)$ experiment. The original 48 constrained-policy intervals, including six favoring FVI and none favoring witness, remain available with newly exposed fee frontiers. Their adverse point-state pattern motivates inspection of actual policies; it is not removed by a favorable all-state certificate.

The complete development edition, unchanged historical companions and machine-readable preservation map retain all of these results and their proofs. This revision adds a controlled resource allocation and actual implementation decision, not a selectively favorable replacement record.

12. COST-DIRECTED CONSTRUCTION AND IMPLEMENTATION EVIDENCE

We now execute the distinction between constructing a safe policy and improving its actual economic performance. The new policies use the same nonlinear investment primitives as Section 11; neither the price parameter nor the terminal penalty is changed to favor the proposed action rule. All earlier continuation weights, owners, policies, clocks and adverse contrasts retain their historical identities. The recovered R50 main experiment and common-accuracy amendment had separate local pre-execution freezes. R51 is an explicitly known-design reproduction, with its own Git source identity and runner clocks; the same draws do not provide additional independent observations.

12.1. Prospective resource selection and a residual-driven comparator

For the primary block, select powers of two $N \in \{4, 8, 16, 32, 64\}$ and $K, M \in \{1, 2, 4, 8, 16\}$ before constructing a policy. Among allocations satisfying the primitive witness budget $a/N + b/K + q/M \leq \varepsilon - 10^{-8}$, minimize the exact integer query proxy $T(N + 1)^d(K + 1)M$, with lexicographic ties. The reserve is checked

against the realized certificate; it is not a universal fixed-precision theorem. This rule supplies an executed version of the separated allocation, rather than a retrospective proxy calculation. Conventional FVI receives the same allocation but its own actual-modulus certificate. The witness planning formula alone does not guarantee FVI attainment.

The catalogue contains seven economic cells: $d = 2, 3, 4$, $T = 2$, $p = 1$, $\varepsilon = 5$, and $d = 2$, $T = 2, 3$, $p = 1, 4$, $\varepsilon = 2$. Witness and uniform FVI are each run on all cells. Two additional services choose the smallest isotropic power-of-two allocation $N = K = M$ satisfying the same primitive target-five budget in $d = 2$, $T = 2$, $p = 1$. A further service in that cell uses residual-driven FVI. Each of these seventeen specifications is executed three times in separate single-CPU processes, with identical warm-up and rotated service order. Repetitions estimate local timing dispersion, not a distribution of economic tasks or neural initializations.

The residual-driven comparator queries its own future Bellman objective at interval endpoints and midpoints. It inserts coordinate knots where the largest interpolation surplus across the other coordinates is greatest, then equidistributes square-root surplus mass along each axis, with dyadic rounding and explicit monotone-knot checks. This last step follows the local quadratic interpolation-error scaling, rather than the curvature-mass rule of the earlier graded block. All pilot queries, residual comparisons, compilation and output are charged. Recorded geometry, not the method name, determines whether adaptation occurred. This is a coordinate-tensor comparator, not a claim to have executed the whole adaptive sparse-grid literature.

The original R50 design and source were frozen locally before their measured execution. Its early bisection-only development test and subsequent equidistribution correction are retained with their original records. R51 freezes the recovered algorithm before a known-design reproduction; it is not presented as a new blind experiment. The present clocks belong only to this execution, and the earlier R49 evidence is unchanged.

TABLE VI: Primary prospective allocations and realized certificates

$d;T;p$	Method	N,K,M	ε	Bound	Pass	Sec.	Sec. range
2;2;1	W	32,2,2	2	1.9713	3/3	0.0847	[0.0841,0.0852]
2;2;1	W I	8,8,8	5	4.6296	3/3	0.0195	[0.0193,0.0200]
2;2;1	W	16,1,1	5	3.9427	3/3	0.0240	[0.0238,0.0268]
2;2;4	W	32,8,1	2	1.9317	3/3	0.1086	[0.1072,0.1105]
2;3;1	W	64,4,2	2	1.7567	3/3	0.5059	[0.5001,0.5075]
2;3;4	W	64,4,2	2	1.9149	3/3	0.5058	[0.4962,0.5095]
3;2;1	W	16,1,1	5	4.0581	3/3	0.4424	[0.4335,0.4610]
4;2;1	W	8,8,2	5	4.8628	3/3	1.8441	[1.8348,1.9009]
2;2;1	A	16,1,1	5	4.2586	3/3	1.0709	[1.0636,1.0941]
2;2;1	F	32,2,2	2	2.1589	0/3	0.0644	[0.0631,0.0646]
2;2;1	F I	8,8,8	5	5.3703	0/3	0.0232	[0.0227,0.0233]
2;2;1	F	16,1,1	5	4.2199	3/3	0.0243	[0.0242,0.0246]
2;2;4	F	32,8,1	2	2.0923	0/3	0.0900	[0.0869,0.0918]
2;3;1	F	64,4,2	2	1.7761	3/3	0.3650	[0.3495,0.3661]
2;3;4	F	64,4,2	2	1.9038	3/3	0.3606	[0.3590,0.3723]
3;2;1	F	16,1,1	5	4.3316	3/3	0.3890	[0.3831,0.4258]
4;2;1	F	8,8,2	5	5.653	0/3	2.4360	[2.4275,2.4645]

W: native witness with equivalent ReLU realization; F: uniform FVI; A: residual-driven FVI; I: isotropic allocation. Each specification has three identical mathematical checkpoints and three distinct clocks. The primitive planning guarantee is witness-specific; FVI uses its actual critic modulus and certificate. Decimal rounding never determines attainment.

The primary block certifies 24 of 24 witness services, 12 of 24 uniform FVI services and 3 of three residual-driven services. These counts include timing repetitions, not independent policy draws. The residual comparator has nonuniform axes in all 3 date models of its repeated checkpoint. Its bound is 4.2586, compared with 4.2199 for uniform FVI at the same quota. Its pilot work is retained.

In the isotropic comparison cell, the planned witness uses 1,156 target innovation evaluations, versus 11,664 for the isotropic service. The latter has fewer state nodes;

query savings are therefore not a claim of less work in every category. The table reports measured complete clocks rather than inferring a speed ratio from this count.

12.2. A separate common-accuracy completion

Inspection of the primary block showed that sharing a witness-sufficient allocation does not make the FVI certificate pass. We therefore froze a separate, explicit service: in $d = 2, 3, 4$ at $T = 2, p = 1, \varepsilon = 5$, begin with the same planner; if a method's actual certificate fails, double only N , reconstruct from primitives and retain the failed attempt. Stop at first success or at $N = 64$. Both methods face this rule. There are three new isolated repetitions per method and dimension. The source freeze and results are separate from the primary catalogue, and old failures remain failures.

TABLE VII: Separate common-target completion: horizon two, price one, target five

d	Method	N range	Tries	Bound	Pass	Sec.	Sec. range	MiB
2	W	16–16	1	3.9427	3/3	0.0242	[0.0223,0.0281]	41.4
2	F	16–16	1	4.2199	3/3	0.0244	[0.0236,0.0275]	41.7
3	W	16–16	1	4.0581	3/3	0.4383	[0.4155,0.4386]	44.3
3	F	16–16	1	4.3316	3/3	0.3913	[0.3840,0.3935]	42.3
4	W	8–8	1	4.8628	3/3	1.7418	[1.7139,1.8030]	50.5
4	F	8–16	2	3.1181	3/3	29.0946	[28.9063,29.3447]	169.8

All failed state refinements are included in the complete prefix. MiB is the maximum process peak over three repetitions, not just model storage. The amendment followed the primary outcomes and was frozen before its own execution. Neither block overwrites the other. Host exclusivity and CPU frequency were not controlled.

The returned policies receive new direct actual-cost comparisons below. These observations provide a common target in every reported dimension. They remain a finite low-dimensional tensor experiment. The construction and corner factors ($N +$

1) ^d and 2 ^d are not removed by the compiler, the terminal repair or this allocation rule.

12.3. Direct policy costs and the source of removable loss

The primary cost experiment fixes twenty-five groups. Sixteen use the original first-target-two R49 witness and FVI policies in all four price-horizon cells, evaluated under the uniform law and each of the three declared point laws. Seven use the new planned services under uniform initial states; one compares residual-driven and uniform FVI; one uses the isotropic allocations. A separate three-group family evaluates the common-accuracy amendment's returned policies. Every group contains both original and final-repaired policies, with the same eight-bit observation and 1/4096 action contract. Both generators receive the same improvement.

Each group uses 131,072 continuous-law interval paths and analytic final-innovation integration. Four absolute costs and all six pair contrasts are recorded. The primary 250 estimands and supplemental 30 estimands are covered by the conservative endpoint union-bound budget of 512 and log upper bound 13. The supplemental groups have their own conditional independent-bin model within the declared simultaneous family; the policies may depend on earlier construction information. Fixed streams establish reproducibility only. The older R49 family retains its own separate error account.

TABLE VIII: Actual policy differences and removable final-action loss

Case	$J_L - J_R$	$J_{L^+} - J_{R^+}$	$J_L - J_{L^+}$
C2;2;1;U	[0.023033,0.028988]	[0.012051,0.017775]	[0.011448,0.014567]
C3;2;1;U	[0.01796,0.023776]	[0.010129,0.015794]	[0.0077535,0.010757]
C4;2;1;U	[0.033636,0.039473]	[0.015912,0.021433]	[0.016432,0.019542]
I2;2;1;U	[0.042713,0.048868]	[0.018209,0.023856]	[0.023541,0.026849]
O2;2;1;M	[0.0064146,0.011415]	[-0.0023843,0.0023843]	[0.007622,0.010239]
O2;2;1;L	[0.054972,0.060123]	[0.044007,0.048792]	[0.0099411,0.012722]
O2;2;1;H	[0.0042173,0.0090667]	[0.0022567,0.007049]	[0.0014934,0.003946]

Case	$J_L - J_R$	$J_{L^+} - J_{R^+}$	$J_L - J_{L^+}$
O2;2;1;U	[0.013161,0.018416]	[0.0064562,0.011605]	[0.0056474,0.0083165]
O2;2;4;M	[0.035287,0.040496]	[0.030381,0.035526]	[0.0036933,0.0063836]
O2;2;4;L	[0.020345,0.025506]	[0.017976,0.023095]	[0.0010828,0.0036974]
O2;2;4;H	[0.039236,0.044497]	[0.034549,0.039692]	[0.0034416,0.006155]
O2;2;4;U	[0.011181,0.016658]	[0.0058295,0.011256]	[0.004188,0.0069254]
O2;3;1;M	[0.015697,0.021868]	[0.012485,0.018641]	[0.0016956,0.0047777]
O2;3;1;L	[0.026367,0.032516]	[0.022928,0.029055]	[0.0020031,0.005116]
O2;3;1;H	[0.011221,0.017268]	[0.0095515,0.015586]	[0.00030976,0.0033545]
O2;3;1;U	[0.0095127,0.015773]	[0.0065302,0.012761]	[0.0014967,0.0045926]
O2;3;4;M	[0.010433,0.016948]	[0.0089337,0.015429]	[9.4671e-05,0.0033779]
O2;3;4;L	[0.0095446,0.01608]	[0.0074164,0.013916]	[0.0005048,0.0037993]
O2;3;4;H	[0.0073415,0.013898]	[0.0046083,0.011119]	[0.0011154,0.0044219]
O2;3;4;U	[0.0056055,0.012273]	[0.0039105,0.010565]	[0.00021625,0.0035063]
Q2;2;1;U	[0.01186,0.017187]	[0.0056983,0.010907]	[0.0055893,0.0083122]
P2;2;1;U	[0.022852,0.028801]	[0.011835,0.01755]	[0.011497,0.014615]
Q2;2;4;U	[0.0086864,0.014074]	[0.004022,0.0093656]	[0.003535,0.0062382]
Q2;3;1;U	[0.01136,0.017753]	[0.0081953,0.014553]	[0.0017403,0.0048791]
Q2;3;4;U	[0.010559,0.017349]	[0.0074835,0.014248]	[0.0020091,0.0053568]
P3;2;1;U	[0.017874,0.023687]	[0.0099988,0.015657]	[0.0078066,0.010814]
P4;2;1;U	[0.033196,0.039029]	[0.015676,0.021193]	[0.016374,0.019482]
A2;2;1;U	[-0.0024517,0.0023493]	[-0.0024125,0.0023985]	[0.00062241,0.0030972]

A case is block/ dimension; horizon; price; initial law. O: original R49 target-two policies; P: primary target-five plan; Q: primary target-two plan; C: common-accuracy amendment; I: isotropic; A: surplus FVI versus uniform FVI. Except A, left is witness and right is uniform FVI. U is uniform; L,M,H are point states 1/8,1/2,7/8. Superscript + denotes the identical terminal safety rule. Five-significant-digit display is not an exact endpoint; the rational records govern all conclusions.

All 27 declared witness comparison groups identify a strictly positive reduction of the witness incumbent's actual expected cost. The simultaneous gain lower endpoints range from 9.4671e-05 to 0.023541; upper endpoints range from 0.0033545 to 0.026849. This is a law-specific statement for the stated policies, not strict improvement at every initial state or 27 independent training successes.

Before repair, witness cost is identified as higher in 27 groups. After both methods receive the same repair, it remains higher in 26, is lower in 0, and is unresolved in 1. The residual-driven FVI comparison is unresolved both before and after repair. Thus the terminal construction repairs a genuine economic loss without manufacturing a reversal of the strong conventional comparison.

The active evaluator records 3,670,016 interval paths, 7,340,032 cell-gate queries and 5,447,560 accepted changes across the two incumbents. No acquisition ambiguity or broad-hull fallback is recorded. Its joint direct-evaluation time before durable record output is 127.95 seconds. That total is evaluation work, not a speed comparison between independently executed policies.

For each incumbent the direct reduction is the centered final contrast. The remaining witness–FVI difference after both repairs is an actual comparison of the two changed policies. The identity

$$J_W - J_F = (J_W - J_{W+}) + (J_{W+} - J_{F+}) - (J_F - J_{F+})$$

separates removable final-action losses from the difference that remains. It does not attribute every remaining discrepancy to a single representation mechanism. The deposited date-level action means, terminal state means, accepted and blocked proposals, and capacity/quantum repair counts permit this distinction to be checked. A finite-sample positive gain can justify a strictly positive replacement charge up to its simultaneous lower endpoint; a deterministic nonincrease alone does not justify an arbitrary implementation fee.

12.4. *Actual-cost frontiers and complete resource boundaries*

The supplement reports every group’s four absolute cost intervals and every direct contrast. For each new construction, its resource coordinate includes the full own-future service and, separately, the joint evaluation charge. The reader can use either a shared fixed evaluation cost for the catalogue decision or the complete joint cost required to certify a particular comparison. We do not claim to have measured independent marginal costs for four separately simulated policies when

the implementation shares their earlier trajectories. The pair clock is an explicitly charged object, not an unpriced convenience.

TABLE IX: Construction work and actual cost of new implementations

Case	Method	Build sec.	Original cost	Repaired cost	Pair sec.
C2;2;1;U	W	0.0242	[0.6242,0.6378]	[0.61117,0.62482]	3.799
C2;2;1;U	F	0.0244	[0.59821,0.61177]	[0.59628,0.60988]	3.799
C3;2;1;U	W	0.4383	[0.58992,0.60068]	[0.58062,0.59146]	6.102
C3;2;1;U	F	0.3913	[0.56904,0.57982]	[0.56768,0.57849]	6.102
C4;2;1;U	W	1.7418	[0.62188,0.63199]	[0.60399,0.61391]	9.790
C4;2;1;U	F	29.0946	[0.5855,0.59525]	[0.5854,0.59515]	9.790
I2;2;1;U	W	0.0195	[0.63959,0.65386]	[0.61456,0.6285]	3.744
I2;2;1;U	F	0.0232	[0.59407,0.60781]	[0.59363,0.60737]	3.744
Q2;2;1;U	W	0.0847	[0.60972,0.62332]	[0.60276,0.61638]	3.870
Q2;2;1;U	F	0.0644	[0.59518,0.60882]	[0.59444,0.6081]	3.870
P2;2;1;U	W	0.0240	[0.6222,0.63577]	[0.60912,0.62274]	3.901
P2;2;1;U	F	0.0243	[0.59639,0.60993]	[0.59445,0.60803]	3.901
Q2;2;4;U	W	0.1086	[0.69243,0.70702]	[0.68754,0.70213]	3.919
Q2;2;4;U	F	0.0900	[0.68106,0.69562]	[0.68086,0.69542]	3.919
Q2;3;1;U	W	0.5059	[0.72237,0.73696]	[0.71906,0.73365]	4.406
Q2;3;1;U	F	0.3650	[0.70782,0.7224]	[0.70769,0.72228]	4.406
Q2;3;4;U	W	0.5058	[0.86908,0.88468]	[0.86539,0.881]	4.513
Q2;3;4;U	F	0.3606	[0.85511,0.87074]	[0.85451,0.87015]	4.513
P3;2;1;U	W	0.4424	[0.58981,0.60055]	[0.58046,0.59128]	6.294
P3;2;1;U	F	0.3890	[0.56902,0.57978]	[0.56764,0.57844]	6.294
P4;2;1;U	W	1.8441	[0.61951,0.62957]	[0.60168,0.61155]	7.713
P4;2;1;U	F	2.4360	[0.58357,0.59328]	[0.58332,0.59303]	7.713
A2;2;1;U	A	1.0709	[0.59772,0.61129]	[0.59585,0.60945]	3.797
A2;2;1;U	F	0.0243	[0.59777,0.61134]	[0.59585,0.60946]	3.797

Pair sec. is a shared cost of the whole four-policy comparison, repeated in the display for clarity, not independently attributable to each row. Construction medians use the same cell and fresh records; no

historical clocks are spliced. Absolute-cost intervals are wider than centered pair intervals. Charges enter Proposition 22 with their explicitly chosen shared or marginal accounting convention.

All stored coefficients, original owners, action scalars, model evaluation categories, transform relaxations, interpolation/corner operations, terminal derivative queries, whole-cell tests, exact fallbacks, numerical interval widths, serialized bytes and process peak resident memory remain in the records. The counters are primitive categories, not a complete FLOP or bit-operation count. The latter would require an additional cost model. Direct comparison time is not added to clocks from another host or to a historical failure in order to manufacture a new complete service.

The retained sensor catalogue has a structural interpretation through equation (47). Its prices remain normalized primitives, and its confidence-envelope regret is relative to the best observed implementation in the finite catalogue. This revision neither estimates a sensor technology nor substitutes a sample-mean winner for the true expected-net-cost minimizer.

12.5. *Historical findings that the new evidence does not replace*

The earlier 93 of 96 strictly higher witness-cost contrasts remain visible in Table IV. Original curvature-FVI never changed its grid in the realized block. The later genuinely graded method also remains an adverse conventional-comparator experiment: it was less precise in all twelve matched final services and slower in all twelve. These two findings concern the two specified heuristics, not all adaptive dynamic programming. The present surplus comparator adds an independently recorded realized partition, while uniform FVI remains the strong baseline.

Likewise, a successful final-action repair does not establish better earlier continuation learning, universal neural superiority, a guarantee for unrestricted non-convex training, or a dimension-free policy construction. What is established is a cost-directed feasible improvement with an explicit safety condition, its executed original-economy specialization, and source-bound resource and actual-cost comparisons. These results extend the NBO policy-evaluation and improvement program without replacing its economic subject or erasing its previous evidence.

13. EXHAUSTIVE ROBUSTNESS IN THE ORIGINAL INVESTMENT ECONOMY

The action-null result permits a diagnostic in which the true policy and its economic cost are held fixed while a critic's irrelevant component changes. This avoids confusing a new representation with a new economic policy. Before execution, we fixed all eight R49 target-two checkpoint policies retained in the R51 package: witness and uniform FVI for every two-state price-horizon cell. We enumerate every one of the 256^2 closed eight-bit observation cells for each policy. There is no selection of cells by favorable cost or by the new results.

For each cell we retain the original robust capacity repair, the action lattice $1/4096$, and the convex final-date proposal. The centered outward interval for its actual conditional cost difference is the same object used in the inherited terminal gate. We compare the global-width and contrast-aware gates after replacing g by $g - n_M$, for $M \in \{0, 1, 16, 4096\}$. Proposition 13 establishes the zero action-error allowance algebraically; it is not estimated from this grid. Closed-cell enclosures cover boundary states, while execution retains the original observation tie convention. Symbolic cancellation precedes interval evaluation.

TABLE X

EXHAUSTIVE ACTION-NULL ROBUSTNESS ON THE ORIGINAL INVESTMENT ECONOMY

Incumbent	T	p	$M = 0$	$M = 1$	$M = 16$	$M = 4096$
Witness	2	1	48872	0	0	0
Witness	2	4	60574	0	0	0
Witness	3	1	48634	0	0	0
Witness	3	4	60114	0	0	0
FVI	2	1	42198	0	0	0
FVI	2	4	54249	0	0	0
FVI	3	1	37574	0	0	0
FVI	3	4	54249	0	0	0

Each row enumerates all 65,536 closed eight-bit cells. Entries count strict changes accepted by the global-width gate. The contrast-aware gate accepts exactly the $M = 0$ column at every amplitude, with identical returned actions. Earlier policy dates are unchanged. Counts concern cells, not additional expected-cost observations.

The records retain the incumbent and proposed action indices, the returned contrast-aware index, the exact binary64 endpoint encodings, and every global-width decision for every cell. An independent record audit checks cell completeness, cell-wide capacity, all gates, counts and accepted upper bounds. The contrast-aware actions are also compared with a fresh call to the unmodified inherited terminal implementation on every cell. Thus the reported invariance concerns the actual stored policy, not equality of two rounded average costs. The exact finite-state suite separately exercises nonzero action-error bounds, rejection, and all passes of the finite-sweep theorem.

The experiment measures resistance to irrelevant continuation error. It is not a new estimate of $J^{\text{witness}} - J^{\text{FVI}}$. The contrast-aware policies are identical to the unperturbed terminal-repaired incumbents, so repeating their cost simulation would not create additional independent economic observations. The prior actual-cost intervals, their confidence events, and the 93 witness-higher, zero witness-lower and three unresolved R49 comparisons remain unchanged. Safe improvement of either incumbent remains different from superiority over the competing policy.

Every service includes input loading, proposals, true-cost interval comparisons, independent inherited-policy replay, amplitude decisions and durable cell output. Machine-independent query counters, serialization time and peak memory are retained separately. The supplementary work table is the cost of this exhaustive diagnostic, not the construction cost of an alternative learned model. A full T -pass investment-economy execution is not inferred from a final-date ablation. The general finite-sweep result has explicit verified-error hypotheses; the continuous economic experiment executes only the declared specialization.

14. FULL POLICY SWEEPS IN THE NONLINEAR INVESTMENT ECONOMY

The directed certificate is now evaluated beyond an exact-null identity and beyond the final decision. This section uses the original continuous-law two-state investment economy, with state-dependent capacity, nonlinear costs and transitions, and acquired-state implementation. It does not substitute a finite-state model for that economy. The construction and direct-cost records were executed in the separately frozen R53 primary and adaptive studies; the present revision integrates

1 them into the paper and replays their full deposited outputs. That replay adds 1
 2 neither new policies nor new independent samples. 2
 3 3
 4 4

5 14.1. *Design and the economic endpoint* 5 6 6

7 The primary design has horizons $T = 2, 3$, price parameter $p = 1$, and two genera- 7
 8 tors: the compiled witness construction and tensor FVI. Each service starts from the 8
 9 primitives and executes every rung of the original prefix up to the declared check- 9
 10 point. It checks that its reconstructed original policy has the same identity as the 10
 11 historical checkpoint. A five-bit observation contract in each coordinate then gives 11
 12 32^2 closed cells per date, with downward action spacing $1/4096$. Both methods use 12
 13 the same acquisition and robust-feasibility rules. 13

14 Starting from that complete acquired incumbent, each service executes exactly 14
 15 T all-date passes. At every pass all new dates are computed against the same old 15
 16 complete incumbent and then adopted together. Each cell has nine robustly feasible 16
 17 proposed actions and eight interval covers of the full continuous-action range. 17
 18 Sixteen dyadic common-innovation bins enclose every unresolved expectation; the 18
 19 last innovation is integrated analytically. Rectangle extrema enclose discontinuous 19
 20 future actors. These calculations implement the directed certificate of Section 7 with 20
 21 the verification gauge $h = 0$. They are not a claim that a zero critic generates an 21
 22 effective policy, and their policy-evaluation cost is recorded. 22

23 The primary numerical endpoint is the actual expected cost of each complete 23
 24 returned policy under the original uniform initial law and continuous innovations. 24
 25 Conditional on the frozen policies, 65,536 common-path rows per horizon produce 25
 26 outward path-cost enclosures using forty-bit continuous-bin cells. The bounded 26
 27 empirical-Bernstein account applies to the endpoint samples under the declared 27
 28 independent-bin model. Dependence between policies through common paths is 28
 29 retained. The primary protocol allocates a simultaneous family error of $1/100$; the 29
 30 adaptive extension has a separate $1/100$ family. Combining the two families requires 30
 31 the corresponding union bound, rather than counting their repeated policies as 31
 32 independent economic objects. 32

TABLE XI

ACTUAL EXPECTED POLICY COSTS BEFORE AND AFTER FULL SWEEPS

T	Method	Initial cost	Final cost	Initial minus final
2	W	[0.60258, 0.62351]	[0.59092, 0.61168]	[0.00665, 0.01684]
2	F	[0.58999, 0.61078]	[0.58984, 0.61063]	[-0.00463, 0.00494]
3	W	[0.71853, 0.74116]	[0.70898, 0.73147]	[0.00344, 0.01579]
3	F	[0.70576, 0.72833]	[0.70576, 0.72833]	[-0.00596, 0.00597]

W is the compiled witness construction; F is tensor FVI. Every cost is for a complete acquired policy under the original continuous initial and innovation laws. Endpoints are rounded outward for display.

Paired gain intervals, not subtraction of the displayed marginal intervals, determine gain signs. The policies and observations were frozen and executed in the R53 primary study.

TABLE XII

PAIRED WITNESS-MINUS-FVI COST AFTER EVERY PASS

T	Pass	Expected-cost difference
2	0	[0.00757, 0.01775]
2	1	[0.00151, 0.01148]
2	2	[-0.00376, 0.00590]
3	0	[0.00663, 0.01896]
3	1	[0.00285, 0.01513]
3	2	[-0.00105, 0.01112]
3	3	[-0.00288, 0.00924]

A positive interval identifies higher witness cost. An interval containing zero is unresolved, not evidence of equality or witness superiority. The same paths couple policies within a horizon; no independence between these estimands is assumed.

The distinction between cumulative gain and comparative ranking remains important. In the three-date experiment, the witness policy's gain after three full sweeps lies in $[0.00344390, 0.01578829]$, with the displayed endpoints rounded outward. Its final witness-minus-FVI interval is $[-0.00287355, 0.00923410]$. Thus the cumulative own-incumbent gain is identified, while the final cross-method difference is unresolved. The latter is not evidence of equality and not an identified

reversal of the original ranking. Individual step gains can be statistically unresolved even though the deterministic certificate guarantees nonworsening expected cost at every true state.

For an installed witness controller with a replacement charge $\tau \geq 0$, let $[l, u]$ be the corresponding cumulative gain interval. The net benefit of replacement is in $[l - \tau, u - \tau]$ on the same simultaneous event. Hence every $0 \leq \tau < l$ is supported as strictly beneficial when $l > 0$, and every $\tau > u$ is ruled out as beneficial. The intermediate region is unresolved. This gives a full adoption-fee account for an actual improved policy, rather than only a reduced certificate for an unchanged policy. These are normalized model units, not an estimated installation charge or a calibrated welfare exercise.

14.2. All-date acceptance and the original Bellman gap

The improvement account records each proposal, the exact retained incumbent, the interval endpoints, the selected action, the continuous-action lower bound and the induced directed gap. Table XIII gives every primary pass, including nonterminal changes. The supplementary tables report every date's positive contrast allowance and directed gap, the number of blocked and admitted candidates, and the comparison with scalar-width and symmetric-contrast gates. No failed or blocked cell is omitted.

There are two distinct guarantees. A nonpositive directed upper advantage gives safety for an accepted action. The lower interval cover of all feasible actions gives the additional accuracy quantity $\varepsilon_t^k = \sup_C (U_t^k - L_t^k)$. The computed recursion is then $G_t^{k+1} = \beta_t G_{t+1}^k + \varepsilon_t^k$, initialized by a valid support bound. Its terms include cell uncertainty, actor ambiguity, action covering, integration and arithmetic through their actual endpoint propagation. They are not separately estimated from residual samples and then presumed negligible.

A coarse continuous-action cover or a wide future-actor range can make G conservative. The theorem guarantees that the returned policy does not worsen, not that a particular numerical upper bound must decrease at every pass. Executing T passes at nonzero tolerances also does not assert exact optimality: the remaining

TABLE XIII

ALL-DATE POLICY CHANGES AND CERTIFIED GAP RECURSION

T	Method	Pass	Changed	Nonterminal	Gap bound	Prefix (s)
2	W	1	590	17	3.72986	3.462
2	W	2	265	265	0.08149	6.608
2	F	1	653	1	3.59553	3.477
2	F	2	0	0	0.06317	6.639
3	W	1	612	44	5.11709	51.769
3	W	2	282	282	3.65096	101.038
3	W	3	135	135	0.22556	149.583
3	F	1	64	0	4.87775	50.846
3	F	2	0	0	3.40006	99.948
3	F	3	0	0	0.09350	149.173

Changed counts refer to distinct deployed cell-date actions in that pass, not to candidate comparisons. The gap is the directed all-state upper bound against the original continuous-action Bellman optimum. It need not decrease monotonically. Prefix work includes primitive construction and previous passes through their recorded array outputs; the final service and inference records are charged separately in the catalogue account.

gap is the displayed sum of propagated directed errors. The exact T -pass optimality statement and its sharp coefficient have their own rational finite-model regression tests.

14.3. *A genuinely nonuniform conventional comparison*

The extension adds surplus-driven FVI at the same two-date primitive checkpoint. Its adaptation history and the number of nonuniform date models are recorded, together with every pilot and every earlier construction rung. It is not the earlier curvature rule that left its grid unchanged. Compiled witness and tensor FVI are reconstructed on the same extension runner, receive the same two complete passes, and reproduce their primary policy identities. A fresh path stream compares all three methods and all passes within a separately frozen statistical family.

TABLE XIV

FRESH ACTUAL-COST COMPARISON WITH A NONUNIFORM CONVENTIONAL METHOD

Method	Pass	Actual expected cost	Gain from initial policy
W	0	[0.60337, 0.62425]	–
W	1	[0.59714, 0.61789]	[0.00129, 0.01130]
W	2	[0.59173, 0.61245]	[0.00662, 0.01681]
F	0	[0.59081, 0.61156]	–
F	1	[0.59065, 0.61141]	[-0.00462, 0.00494]
F	2	[0.59065, 0.61141]	[-0.00462, 0.00494]
A	0	[0.59068, 0.61143]	–
A	1	[0.59060, 0.61135]	[-0.00470, 0.00486]
A	2	[0.59060, 0.61135]	[-0.00470, 0.00486]

A is surplus-driven FVI; W and F denote the same generators as above. All three are reconstructed in the same separately frozen cohort and receive two full improvement passes. W and F policy identities reproduce the primary cohort, while the cost paths are a fresh family. Nonuniformity and pilot work are recorded explicitly.

This comparison tests an actual changed conventional representation, rather than merely a new label for the tensor baseline. Its adaptive grid nevertheless remains a tensor grid with nonuniform coordinates. It is therefore not presented as a sparse-grid or non-tensor high-dimensional demonstration. The earlier failed curvature rule, the adverse graded refinement, and the original direct-cost comparisons remain in their original tables and records.

14.4. Complete catalogue work at common actual-cost targets

A finite catalogue of returned policies supplies a precise work account. For each method we select the pass with the lowest simultaneous actual-cost upper endpoint. We charge that method's entire observed construction-and-improvement service, including all executed passes and durable records, plus the entire shared cost-inference service. Charging the full shared service to each eligible method is conservative and avoids assigning an unobserved marginal inference time to a

method evaluated only jointly. The separate publication audit reports the additional replay and document-production overhead.

TABLE XV
COMPLETE-CATALOGUE WORK TO A VERIFIED ACTUAL-COST UPPER BOUND

Cohort	Method	Pass	Cost upper	Service (s)	Inference (s)	Sum (s)
P2	W	2	0.61168	6.612	1.850	8.463
P2	F	1	0.61063	6.643	1.850	8.493
P3	W	3	0.73147	149.587	3.182	152.769
P3	F	1	0.72833	149.178	3.182	152.359
A2	W	2	0.61245	6.646	2.667	9.312
A2	F	1	0.61141	6.579	2.667	9.245
A2	A	1	0.61135	18.521	2.667	21.188

For each method the selected pass minimizes its actual-cost upper endpoint. Work conservatively charges every own executed construction and sweep, plus the entire shared inference service. It is not a first-crossing or minimum-work sequential stopping estimate. Replay and document-production overhead are separately timed in the release audit. The exact finite threshold partition, eligible methods and least-recorded-work choices are deposited in the machine-readable audit.

For a target u , the eligible methods are exactly those whose selected actual-cost upper bound is at most u . Taking the least recorded complete-catalogue work among eligible methods defines the observed finite frontier. Its breakpoints and method choices are deposited without rounding; the printed table is descriptive. This is a frontier for releasing a policy from an already executed finite catalogue. It is not a claim about the minimum work of an unexecuted sequential stopping algorithm, an optimized resource allocation, or a universal fastest generator. In particular, improving the witness incumbent does not by itself establish a better work-to-cost frontier than FVI.

The work tables separately identify construction time, pair-node counts, coupling work, actor queries, actor-table bytes, process peak memory, and the cumulative clocks. A common-innovation certificate avoids a stored table over all state pairs, but the work bound in Proposition 16 makes its horizon dependence explicit. These

records turn a constructive mathematical certificate into an executed and costed verification method in the stated economy. They do not remove the tensor observation cover or establish a representation-specific scaling advantage.

14.5. *Controlled approximate-null and false-null tests*

The extension varies three departures from the exact action-null example: the ReLU direction, the deterministic action exposure, and a known action-dependent innovation-mean coefficient. Each departure ranges over 0, $1/4096$, $1/256$, and $1/16$; the amplitudes are 1, 16, and 4096. Two original incumbents yield 384 controlled cases, each evaluated on all 32^2 closed cells. A verified positive allowance replaces the zero certificate whenever the altered coefficient is nonzero.

TABLE XVI
CONTROLLED DEVIATIONS FROM EXACT ACTION-NULL STRUCTURE

Quantity	Recorded value
Controlled cases	384
Closed cells per case	1024
Cell–case combinations	393216
Positive-bound cases	378
Safe changed cell–cases	170374
Harmful false-null cell–cases	116185
Maximum contrast allowance	315.00000
New statistical observations	0

The catalogue changes known ReLU directions, action exposures and action-dependent innovation means. Harm means a strictly positive lower true-cost difference for an action selected by the deliberately false-null rule. Corrected decisions pass the nonpositive true-upper check. Cell–cases are deterministic repeated diagnostics, not independent economic observations.

The corrected gate and the deliberately false-null gate are both replayed. Every selected corrected action is feasible and has a nonpositive true-advantage upper endpoint. For the false-null policy, a strictly positive true-advantage lower endpoint identifies a harmful decision, not merely an uncertain one. The full catalogue retains

accepted, blocked and unchanged actions, including cases where large non-nullity makes the verified gate conservative. These are deterministic diagnostics for known perturbed primitives. They are not additional cost-inference samples, an automatic discovery of learned null structure, or a guarantee under an unknown misspecified transition law.

15. ECONOMIC SCOPE AND CONCLUSION

The economic contribution is an explicit connection between a policy's construction resources and its implementable cost. A primitive error budget can determine whether to refine the state cover, action search or integration. A complete finite tolerance frontier states when one generator returns a certified policy earlier. Independent comparisons of the returned policies then determine what can be said about controller replacement and purchased information. These are different steps; none can be replaced by the favorable outcome of another.

The constructive backend and exact compiler make positive deterministic statements under explicit moduli and effective-computability conditions. Their guarantees are not success probabilities for unrelated nonconvex training algorithms. The matched experiments and direct-cost intervals make finite, source-bound claims about the declared economies, resource ladders and initial laws. Common-accuracy dimension cells do not establish an asymptotic scaling law, and endogenous precision within a finite sensor catalogue is not an optimum over all information technologies. The original broad theory, applications and adverse experimental record remain part of the complete NBO development.

The cost-directed result adds a further positive statement: a feasible acquired policy can be improved without sacrificing its incumbent cost guarantee, provided its centered advantage is verified. In the executed final-stage specialization every studied witness incumbent has a strictly positive identified reduction under the declared initial law. Most witness-FVI differences remain unfavorable after both methods are repaired. The improvement theorem and this comparative observation are consistent: safe improvement relative to an incumbent is not dominance over another method.

The action-contrast result further separates continuation representation from decision-relevant error. Its coupled-residual certificate and action-null correction preserve the same safety and optimality targets. The exhaustive cell study checks an exact robustness property on the original economy; its identical returned policies do not supply additional independent cost observations.

Neural Bellman Operators therefore provide a way to organize continuation construction, feasible implementation, certification and economic resource choice around the policy actually returned. Whether a particular representation or generator is preferable is decided by its full certified frontier and actual economic consequences, rather than by the name of its function class.

15.0.0.1. Full-horizon execution and adoption. The new full-sweep calculation changes nonterminal as well as terminal decisions while retaining exact incumbents whenever the whole-cell comparison is inconclusive. Its directed recursion, continuous-action lower covers, and positive contrast bounds are evaluated in the original nonlinear two-state economy. The cost intervals identify cumulative gains for the witness incumbent after complete sweeps. They do not identify a uniform advantage over conventional policies; the adaptive and tensor comparators receive the same improvement and inference treatment. The recorded gain intervals determine the entire range of replacement fees supported by the experiment, in the normalized units of the model.

The paired recursion avoids storing a full state-pair table, but its demonstrated implementation still uses a tensor observation cover and has explicit horizon dependence. The nonuniform conventional experiment is not a non-tensor high-dimensional experiment. These distinctions leave the original NBO research objective intact while keeping each mathematical and empirical assertion tied to its actual assumptions and evidence.

APPENDIX A: SUBMISSION AND PRESERVATION MAP

This article and its technical supplement are the active R54 exposition of the same paper. Every R49 main-article and supplementary label is retained in its corresponding current document. The complete development editions retain the

R48 main and supplementary bodies and add the current theory and evidence, with a reading notice distinguishing historical wording from the active empirical account. The original R49 sources remain unchanged under `preserved/R49`; R48 sources and earlier theory/application companions are likewise retained. Preservation is not a new mathematical validation of every historical application.

The local revision contains ordinary sources, both immutable R49 evidence archives, the R48 publication input, newly executed records, generated tables, a point-by-point response and compiled documents. The direct numerical correction retains its original observations and replays the same streams with explicitly outward cross-policy subtraction. The corrected intervals, not their raw predecessors, govern the active direct-cost tables. Source and evidence identities are checked before a clean offline Git-archive rebuild. The local delivery record distinguishes a completed build from a remote publication: no remote branch or commit is claimed without a confirmed push. This limitation is administrative, not a substitute for the substantive revised manuscript.

APPENDIX: REFERENCES

- FELZENSZWALB, PEDRO F. AND DANIEL P. HUTTENLOCHER (2012): “Distance Transforms of Sampled Functions,” *Theory of Computing*, 8 (19), 415–428. [3, 10]
- FERNS, N., P. PANANGADEN, AND D. PRECUP (2004): “Metrics for Finite Markov Decision Processes,” in *Proceedings of the Twentieth Conference on Uncertainty in Artificial Intelligence*, 162–169. [5]
- GRIMM, C., A. BARRETO, S. SINGH, AND D. SILVER (2020): “The Value Equivalence Principle for Model-Based Reinforcement Learning,” in *Advances in Neural Information Processing Systems*, 33. [5]
- KAKADE, S., AND J. LANGFORD (2002): “Approximately Optimal Approximate Reinforcement Learning,” in *Proceedings of the Nineteenth International Conference on Machine Learning*, 267–274. [5]
- LAKSHMINARAYANAN, CHANDRASHEKAR AND SHALABH BHATNAGAR (2014): “Approximate Dynamic Programming based on Projection onto the $(\min, +)$ Subsemimodule,” arXiv:1403.4175. [3]
- MAURER, ANDREAS AND MASSIMILIANO PONTIL (2009): “Empirical Bernstein Bounds and Sample Variance Penalization,” in *Proceedings of the 22nd Annual Conference on Learning Theory*, arXiv:0907.3740. [3, 32]
- MCSHANE, EDWARD J. (1934): “Extension of Range of Functions,” *Bulletin of the American Mathematical Society*, 40 (12), 837–842. [3, 9]
- METELLI, ALBERTO MARIA, MATTEO PIROTTA, DANIELE CALANDRIELLO, AND MARCELLO RESTELLI (2021): “Safe Policy Iteration: A Monotonically Improving Approximate Policy Iteration Approach,” *Journal of Machine Learning Research*, 22 (97), 1–83. [13]

- MUNOS, R. (2003): “Error Bounds for Approximate Policy Iteration,” in *Proceedings of the Twentieth International Conference on Machine Learning*, 560–567. [5]
- NG, A. Y., D. HARADA, AND S. RUSSELL (1999): “Policy Invariance under Reward Transformations: Theory and Application to Reward Shaping,” in *Proceedings of the Sixteenth International Conference on Machine Learning*, 278–287. [5]
- PIROTTA, MATTEO, MARCELLO RESTELLI, ALESSIO PECORINO, AND DANIELE CALANDRIELLO (2013): “Safe Policy Iteration,” in *Proceedings of the 30th International Conference on Machine Learning*, vol. 28 of *Proceedings of Machine Learning Research*, 307–315. [5]
- WANG, YANG, BRENDAN O’DONOGHUE, AND STEPHEN BOYD (2015): “Approximate Dynamic Programming via Iterated Bellman Inequalities,” *International Journal of Robust and Nonlinear Control*, 25 (10), 1472–1496. [3]